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FLUIDWORLD: Reaction-Diffusion Dynamics as a Predictive Substrate for World Models

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https://github.com/infinition/FluidWorld

Abstract

World models learn to predict future states of an environment, enabling planning and mental simulation. Current approaches default to Transformer-based predictors operating in learned latent spaces. This comes at a cost: $O(N^2)$ computation and no explicit spatial inductive bias. This paper asks a foundational question: *is self-attention necessary for predictive world modeling, or can alternative computational substrates achieve comparable or superior results?* I introduce FLUIDWORLD, a proof-of-concept world model whose predictive dynamics are governed by *partial differential equations* (PDEs) of reaction-diffusion type. Instead of using a separate neural network predictor, the PDE integration *itself* produces the future state prediction. In a strictly parameter-matched three-way ablation on unconditional UCF-101 video prediction (64×64, $\sim$800K parameters, identical encoder, decoder, losses, and data), FLUIDWORLD is compared against both a Transformer baseline (self-attention) and a ConvLSTM baseline (convolutional recurrence). While all three models converge to comparable single-step prediction loss, FLUIDWORLD achieves $2\times$ lower reconstruction error, produces representations with 10–15% higher spatial structure preservation and 18–25% more effective dimensionality, and critically maintains coherent multi-step rollouts where both baselines degrade rapidly. All experiments were conducted on a single consumer-grade PC (Intel Core i5, NVIDIA RTX 4070 Ti), without any large-scale compute. These results establish that PDE-based dynamics, which natively provide $O(N)$ spatial complexity, adaptive computation, and global spatial coherence through diffusion, are a viable and parameter-efficient alternative to both attention and convolutional recurrence for world modeling.

1 Introduction

Learning predictive models of the world, commonly called *world models*, is a central challenge in artificial intelligence [18, 12]. A world model takes an observation x_t and predicts the future state x_{t+1} or its abstract representation, optionally conditioned on actions. Such models enable agents to plan by simulating the consequences of candidate actions before execution [15, 30].

The dominant paradigm uses *Transformer*-based architectures [33] as the predictive engine. But why? This is a choice by default, not by principle. LeCun's Joint Embedding Predictive Architecture (JEPA) [18], implemented in I-JEPA [1] and V-JEPA [2], predicts latent representations rather than pixels, using Vision Transformers (ViT) [8] as both encoders and predictors. The approach is powerful, yet it has fundamental limitations:

1. $O(N^2)$ **spatial cost.** Self-attention over N spatial tokens scales quadratically, limiting resolution.
2. **No spatial inductive bias.** Transformers must *learn* spatial propagation from data, consuming model capacity for what physics provides for free.

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

3. **Fixed computation.** Every prediction costs the same, regardless of complexity.
4. **No persistent state.** Each prediction is independent; temporal context requires explicit memory mechanisms.

This paper is an *architectural proof-of-concept*. I do not aim to beat state-of-the-art world models that use orders of magnitude more parameters and compute. The question is narrower: is attention strictly necessary for predictive world modeling? Or can an alternative substrate, one grounded in physics rather than combinatorics, match or exceed Transformers at equal parameter budget?

World models are ultimately designed for action-conditioned planning, but their foundational prerequisite is *stable temporal prediction*. The experiments here focus entirely on unconditional video prediction, to isolate the predictive capacity of the PDE substrate. The FLUIDWORLD architecture does natively support action conditioning via additive forcing terms in the PDE, but I have not yet evaluated that capability. It remains the most important next step.

I propose FLUIDWORLD, a world model that replaces attention-based prediction with *reaction-diffusion partial differential equations* (PDEs). The key insight is simple: the PDE integration itself *is* the prediction. Encode an observation into a spatial feature map; let diffusion propagate spatial information, learned reaction terms handle nonlinear transformation, and optional forcing terms condition on actions. The latent state evolves toward the predicted future. What falls out naturally from this formulation is $O(N)$ local computation, adaptive convergence, and continuous temporal dynamics.

The contributions of this work are:

1. **PDE-native world model.** I demonstrate that reaction-diffusion dynamics can serve as the predictive engine of a world model, replacing self-attention entirely (§4).
2. **BeliefField.** A persistent latent state that accumulates temporal context through PDE evolution, with biologically-inspired mechanisms (Hebbian diffusion, synaptic fatigue, lateral inhibition) that improve representational diversity (§4.4).
3. **Three-way parameter-controlled ablation.** I compare FLUIDWORLD against both a Transformer baseline (self-attention) and a ConvLSTM baseline (convolutional recurrence) at *identical* parameter count ($\sim$800K), same encoder, same decoder, same losses, and same data, isolating the effect of the predictive substrate (§5).
4. **Efficiency and rollout analysis.** I characterize the $O(N)$ vs $O(N^2)$ scaling advantage, show that PDE dynamics produce richer spatial representations per parameter, and demonstrate superior multi-step rollout coherence compared to both baselines (§6).

2 Related Work

World Models. Ha and Schmidhuber [12] introduced the concept of learned world models using VAE encoders and RNN-based dynamics. Dreamer [13, 14, 15] extended this with RSSM (Recurrent State-Space Models), achieving strong results in continuous control. MuZero [30] learns a world model for planning in discrete action spaces. IRIS [25] and TWM [28] use Transformer-based world models with discrete latent spaces. All these approaches use standard neural network architectures (RNNs, Transformers, MLPs) as their predictive engine.

JEPA and Self-Supervised Prediction. LeCun [18] proposed the Joint Embedding Predictive Architecture (JEPA) as a blueprint for autonomous intelligence, predicting in representation space rather than pixel space. I-JEPA [1] and V-JEPA [2] validate this framework for images and video using ViT predictors. MC-JEPA [4] separates motion and content. These works establish representation prediction as viable but retain Transformer predictors. Concurrently, Qu *et al.* [27] demonstrate that JEPA with VICReg regularization learns more physically informative representations than pixel-level methods (MAE, autoregressive models) on PDE-governed spatiotemporal systems, further validating latent prediction as the right learning paradigm for physical dynamics.

PDE-Inspired Neural Networks. Neural ODEs [7] interpret residual networks as ODE discretizations. PDE-Net [22, 21] learns PDE coefficients for physical simulation. Neural Operators [19] learn

solution operators for PDEs. Reaction-diffusion networks have been explored for image segmentation [24] and graph processing [6]. This work differs fundamentally. I do not use PDEs to *simulate* physical systems. I use them as the *computational substrate* of a learned world model. The PDE is not the thing being modeled. It is the model.

Video Prediction. Convolutional recurrent architectures have a long history in video prediction. ConvLSTM [31] introduced convolutional gates for spatiotemporal sequence modeling. PredRNN [34, 35] proposed spatiotemporal LSTM units with zigzag memory flow. SimVP [9] showed that simple convolutional architectures can match recurrent models on standard benchmarks. These approaches provide a natural middle ground between purely spatial (ConvNet) and purely global (Transformer) processing, and represent important baselines for any spatial prediction architecture.

Efficient Alternatives to Attention. Linear attention [16], state-space models (S4, Mamba) [11, 10], and local attention patterns [20] reduce the $O(N^2)$ cost. My approach is orthogonal. Rather than making attention cheaper, I replace it entirely with PDE dynamics that are $O(N)$ by construction.

3 From Language to World Models: The Fluid Architecture Lineage

FLUIDWORLD is the third iteration of a research program exploring reaction-diffusion PDEs as a general-purpose computational substrate, progressively replacing attention in different modalities:

FluidLM (2024): Language. The initial exploration replaced self-attention with reaction-diffusion dynamics in a language model. A 1D Laplacian operator propagated information along the token sequence, with learned reaction terms providing nonlinear mixing. This proof-of-concept demonstrated that PDE dynamics could process sequential data, though at a performance gap compared to standard Transformers for language tasks.

FluidVLA (2025): Vision and Robotics. The architecture was adapted to 2D spatial data using `FluidLayer2D` with a 2D Laplacian operator. FluidVLA achieved competitive results in image classification and real-time robotic control (40 ms inference on an NVIDIA RTX 4070), demonstrating that PDE-based processing scales to visual perception tasks. Key innovations at this stage included multi-scale dilated Laplacians (dilations $\{1, 4, 16\}$), adaptive early stopping based on convergence monitoring, and `MemoryPump` for $O(1)$ global context. FluidVLA was further extended to `FluidLayer3D` for volumetric medical imaging (CT/MRI segmentation), where the $O(N)$ scaling advantage over attention became particularly relevant for processing high-resolution 3D volumes.

FluidWorld (2026): World Models. This work extends the PDE substrate from perception to prediction. The core innovation is that the same reaction-diffusion equation used for spatial encoding also serves as the temporal prediction mechanism, via the BeliefField (§4.4). Biologically-inspired mechanisms (synaptic fatigue, lateral inhibition, Hebbian diffusion) were added to improve representational diversity in the temporal prediction setting, where channel collapse is a greater risk than in single-frame classification.

The same core equation (diffusion plus learned reaction) has now been applied to 1D sequences, 2D images, 3D volumes, and temporal prediction. Each iteration refined the details, but the $O(N)$ complexity and adaptive computation carried through unchanged.

4 Method

4.1 Overview

FLUIDWORLD processes video frames through three stages (Figure 1):

1. **Encode**: A frame $x_t \in \mathbb{R}^{C \times H \times W}$ is mapped to spatial features $z_t \in \mathbb{R}^{d \times H_f \times W_f}$ via patch embedding followed by PDE-based processing layers.
2. **Evolve**: The persistent BeliefField state s_t integrates z_t and evolves through internal PDE dynamics, conditioned on actions, to produce a predicted latent state $\hat{z}_{t+1}$.

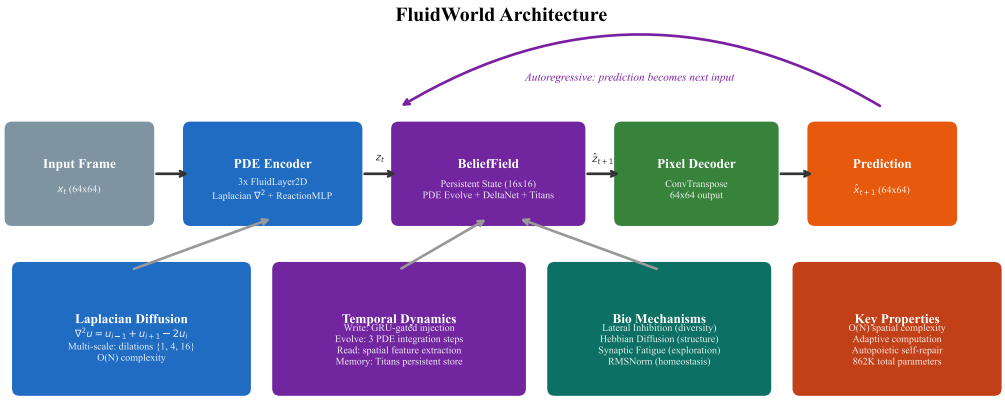


Figure 1: **FLUIDWORLD architecture overview.** *Top row:* The prediction pipeline. An input frame x_t (64×64 pixels) is encoded by three PDE-based layers that use Laplacian diffusion to extract spatial features. These features are written into a persistent *BeliefField*, a 16×16 latent state that accumulates temporal context. The BeliefField evolves via internal PDE dynamics to predict the next state, which the decoder converts back to pixels. During autoregressive rollout, each prediction becomes the next input (purple arrow). *Bottom row:* Key components. The Laplacian kernel ($[1, -2, 1]$) provides spatial information propagation at $O(N)$ cost without attention. The BeliefField combines GRU-gated writing with PDE evolution and Titans persistent memory. Biologically-inspired mechanisms (lateral inhibition, Hebbian diffusion, synaptic fatigue) promote diverse, structured representations. The entire model uses ∼801K parameters (Table 1).

3. **Decode**: A pixel decoder reconstructs frames from latent features for both the current observation (reconstruction) and the predicted future (prediction).

The same PDE equation governs both encoding and temporal evolution. Only the conditioning differs: spatial structure for the encoder, temporal dynamics for the BeliefField. One equation, two roles.

4.2 Reaction-Diffusion Dynamics

Figure 2 illustrates the core mechanism intuitively: Laplacian diffusion spreads concentrated energy until equilibrium, and this same physics smooths away prediction errors during rollout.

The core computation in FLUIDWORLD is the iterative integration of a reaction-diffusion PDE over a spatial feature map $u \in \mathbb{R}^{d \times H_f \times W_f}$:

$$u^{(\tau+1)} = u^{(\tau)} + \Delta t \cdot \Big[\underbrace{D \cdot \nabla^2 u^{(\tau)}}_{\text{diffusion}} + \underbrace{R(u^{(\tau)})}_{\text{reaction}} + \underbrace{\alpha_g \cdot h_g}_{\text{global memory}} + \underbrace{\alpha_l \cdot h_l}_{\text{local memory}} \Big] \tag{1}$$

where τ indexes integration steps (not time steps in the video), Δt is a learned timestep, and:

Diffusion. The Laplacian operator ∇^2 is implemented as a multi-scale discrete convolution using fixed 5-point stencils at multiple dilations $\{d_1, d_2, d_3\} = \{1, 4, 16\}$:

$$D \cdot \nabla^2 u = \sum_k D_k * \text{Conv2D}(u, K_{\text{lap}}, \text{dilation} = d_k) \tag{2}$$

where K_{lap} is the standard discrete Laplacian kernel $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ and $D_k = \text{softplus}(\hat{D}_k) \in \mathbb{R}^d$ are per-channel learned diffusion coefficients. The multi-scale dilations enable information propagation across receptive fields of 3, 9, and 33 pixels in feature space (12, 36, 132 in input space given patch size 4), without any attention mechanism.

Reaction. The reaction term $R : \mathbb{R}^d \to \mathbb{R}^d$ is a position-wise MLP:

$$R(u) = W_2 \cdot \text{GELU}(W_1 \cdot u + b_1) + b_2 \tag{3}$$

with hidden dimension $2d$. This provides per-position nonlinear transformation, analogous to the FFN in Transformers, but without cross-position interaction since diffusion already handles that.

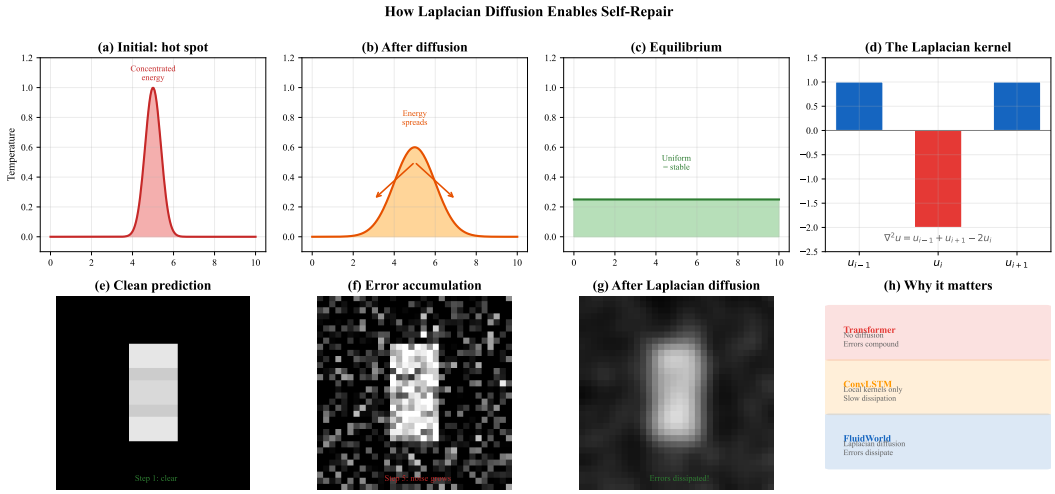


Figure 2: **How Laplacian diffusion enables self-repair.** *Top row (a–d):* The heat equation analogy. A concentrated energy spike (a) spreads via diffusion (b) until reaching equilibrium (c). The Laplacian kernel $[1, -2, 1]$ (d) computes the difference between a position and its neighbors; this single operation drives all spatial propagation in FLUIDWORLD. *Bottom row (e–h):* Application to prediction errors. A clean prediction (e) accumulates noise during autoregressive rollout (f). The Laplacian smooths away high-frequency errors (g), recovering coherent spatial structure. Panel (h) summarizes why this matters: Transformers have no diffusion mechanism (errors compound), ConvLSTMs have only local kernels (slow dissipation), but FLUIDWORLD's Laplacian provides global error correction at every integration step.

Memory terms. Global memory $h_g \in \mathbb{R}^d$ is an $O(1)$ summary accumulated via a gated recurrence:

$$h_g^{(\tau)} = h_g^{(\tau-1)} + \text{sigmoid}(W_{\text{gate}} \cdot \bar{u}) \cdot \tanh(W_{\text{val}} \cdot \bar{u}) \tag{4}$$

where $\bar{u} = \text{mean}(u)$ is the spatial average. Local memory h_l operates at reduced resolution (4×4) and is bilinearly upsampled. Both are broadcast to all spatial positions, modulated by learned coefficients $\alpha_g = \text{softplus}(\hat{\alpha}_g)$ and $\alpha_l = \text{softplus}(\hat{\alpha}_l)$.

Adaptive computation. During inference, integration stops early when the relative change in a low-resolution spatial probe drops below $\epsilon = 0.08$ for `stop_patience` = 2 consecutive steps:

$$\frac{\|u_{\text{probe}}^{(\tau)} - u_{\text{probe}}^{(\tau-1)}\|_1}{\|u_{\text{probe}}^{(\tau-1)}\|_1 + \epsilon'} < \epsilon \tag{5}$$

where u_{probe} is an 8×8 average-pooled version of u. This provides automatic complexity-dependent compute: static scenes converge in $\sim$3 steps, dynamic scenes may use up to 12.

Normalization. RMSNorm [36] is applied every 2 integration steps to stabilize dynamics without eroding the PDE signal at each step.

4.3 Encoder

The encoder maps a frame $x_t \in \mathbb{R}^{C \times H \times W}$ to spatial features $z_t \in \mathbb{R}^{d \times H_f \times W_f}$:

1. **Patch embedding:** A convolutional layer with kernel size p and stride p projects non-overlapping patches: $u_0 = \text{Conv2D}(x_t) \in \mathbb{R}^{d \times H/p \times W/p}$. With $p = 4$ and $H = W = 64$, this yields $u_0 \in \mathbb{R}^{128 \times 16 \times 16}$.
2. **PDE layers:** Three FluidLayer2D modules (Eq. 1) process u_0 sequentially, each running up to `max_steps` integration steps. The layers share the same architectural template but have independent learned parameters (diffusion coefficients, reaction weights, memory gates).

3. **Spatial skip connection:** The encoder output is $z_t = u_{\text{pde}} + u_0$, preserving high-frequency spatial details that PDE diffusion might smooth.
4. **Bio-inspired regularization:** Lateral inhibition and synaptic fatigue (§4.5) are applied to z_t.

4.4 BeliefField: Persistent Temporal State

The BeliefField maintains a persistent spatial state $s_t \in \mathbb{R}^{d \times H_f \times W_f}$ that accumulates temporal context across frames. It operates through three mechanisms:

Write. When a new observation z_t is encoded, it is integrated into the state via a GRU-inspired gate:

$$s_t = \gamma \cdot s_{t-1} + \text{sigmoid}(W_g \cdot z_t) \cdot \tanh(W_v \cdot z_t) \tag{6}$$

where $\gamma = \exp(\hat{\gamma}) \in [0.5, 0.99]$ is a learned decay factor that controls the forgetting rate.

Evolve. The state undergoes internal PDE evolution (Eq. 1) for n_{evolve} steps. This is the core prediction mechanism: the PDE dynamics transform the current state toward the predicted future. The architecture supports optional action conditioning via an additive forcing term in the PDE, though the experiments in this paper evaluate unconditional video prediction.

Read. The predicted next-frame features $\hat{z}_{t+1}$ are extracted from the evolved state via bilinear interpolation to the target spatial resolution: $\hat{z}_{t+1} = \text{Interp}(s_t, H_f \times W_f)$.

4.5 Biologically-Inspired Mechanisms

Temporal prediction is prone to channel collapse, where a few channels dominate and the rest go silent. I introduce three mechanisms, borrowed from neuroscience, to counter this:

Lateral inhibition. Inspired by retinal processing, strong activations suppress weaker ones within each spatial position:

$$\hat{z}_c = z_c \cdot \Big(1 - \beta \cdot \big(1 - \frac{|z_c|}{\max_c |z_c| + \epsilon}\big)\Big) \tag{7}$$

with strength $\beta = 0.3$ and minimum factor 0.2. This encourages sparse, discriminative features.

Synaptic fatigue. Persistently active channels are attenuated proportionally to their cumulative activation:

$$h_c \leftarrow h_c - \kappa \cdot |\bar{z}_c| + \rho, \quad \hat{z}_c = z_c \cdot \text{clamp}(h_c, h_{\min}, 1) \tag{8}$$

where h_c is a per-channel health buffer, $\kappa = 0.1$ is the fatigue cost, $\rho = 0.02$ the recovery rate, and $h_{\min} = 0.1$. This prevents channel collapse by penalizing monopolistic activations.

Hebbian diffusion. Co-activated spatial neighbors strengthen their diffusion pathways:

$$M^{(\tau)} = \lambda M^{(\tau-1)} + \eta \cdot \text{clamp}(u \odot \overline{u}, 0, \infty) \tag{9}$$

where $\overline{u}$ is spatially smoothed via average pooling, $\lambda = 0.99$ is the decay, and $\eta = 0.01$ the learning rate. The Hebbian map M modulates diffusion: $D_{\text{eff}} = D \cdot (1 + \alpha_H \cdot M)$ with gain $\alpha_H = 0.5$. Frequently co-activated pathways diffuse faster, implementing a form of structural plasticity.

4.6 Decoder and Training Objective

Decoder. The pixel decoder maps features back to image space via a symmetric architecture: 1×1 projection ($d \to 64$), two upsampling stages ($\times 2$ bilinear + 3×3 Conv + ResBlock each), and a final 3×3 convolution to output channels. Residual blocks with GroupNorm provide spatial detail reconstruction. The decoder outputs logits; sigmoid is applied in the loss.

Training objective. The total loss combines reconstruction anchoring and predictive objectives:

$$\mathcal{L} = \underbrace{w_r \cdot \mathcal{L}_{\text{recon}}}_{\text{reconstruction}} + \underbrace{w_p \cdot \mathcal{L}_{\text{pred}}}_{\text{prediction}} + \underbrace{w_v \cdot \mathcal{L}_{\text{var}}}_{\text{variance}} + \underbrace{w_g \cdot \mathcal{L}_{\text{grad}}}_{\text{gradient}} \tag{10}$$

where:

- $\mathcal{L}_{\text{recon}} = \text{MSE}(\text{sigmoid}(\hat{x}_t), x_t)$ anchors the encoder to preserve input information.
- $\mathcal{L}_{\text{pred}} = \text{MSE}(\text{sigmoid}(\hat{x}_{t+1}), x_{t+1})$ is the world model prediction objective.
- $\mathcal{L}_{\text{var}} = \frac{1}{d}\sum_c \max(0, \sigma_{\text{target}} - \text{std}(z_c))$ prevents dimensional collapse [3] by encouraging each channel to maintain minimum variance.
- $\mathcal{L}_{\text{grad}} = \frac{1}{2}(\ell_{\text{grad}}(\hat{x}_t, x_t) + \ell_{\text{grad}}(\hat{x}_{t+1}, x_{t+1}))$ with $\ell_{\text{grad}}(a, b) = \|a_x - b_x\|_1 + \|a_y - b_y\|_1$ and subscripts denoting finite-difference spatial gradients. This edge-preservation loss prevents the "predict mean color" collapse mode.

Temporal training. I use truncated backpropagation through time (TBPTT) with window size $T = 4$. For each window, the model processes frames sequentially, accumulating gradients through the BeliefField state. The optimizer step occurs after the full window.

Optimization. AdamW [23] with learning rate 3×10^{-4}, weight decay 0.04, warmup for 500 steps followed by cosine annealing. Gradient clipping at norm 1.0. Mixed precision (FP16) on GPU.

5 Scaling Experiment: PDE vs Transformer vs ConvLSTM

5.1 Experimental Setup

To isolate the effect of the predictive substrate, I design a controlled three-way ablation study with the following constraints:

- **Identical encoder front-end.** All three models use the same PatchEmbed layer (Conv2D, kernel 4, stride 4, $C \to 128$).
- **Identical decoder.** All use the same PixelDecoder architecture (ResBlocks + bilinear upsampling, ∼231K parameters).
- **Identical losses.** All optimize Eq. 10 with the same weights ($w_r = w_p = 1.0$, $w_g = 1.0$, $w_v = 0.5$).
- **Identical data.** UCF-101 [32] at 64×64 resolution, 101 action classes, random temporal crops of $T+1$ frames.
- **Identical training.** Same optimizer (AdamW), same LR schedule, same batch size (16), same number of gradient steps (8,000).
- **Matched parameters.** All three models have ∼800K total parameters (±0.15%).

The *only* difference is the computational engine between encoding and decoding. The three substrates represent fundamentally different inductive biases: PDE dynamics (local diffusion + learned reaction), self-attention (global pairwise interactions), and convolutional recurrence (local spatial gates + LSTM memory).

Hardware. All experiments were conducted on a single consumer-grade desktop: Intel Core i5 CPU, NVIDIA GeForce RTX 4070 Ti (16 GB VRAM), 32 GB RAM. No multi-GPU training, no cloud compute, no distributed data parallelism. Total training time for each model (8,000 steps): approximately 17 minutes for ConvLSTM, approximately 26 minutes for the Transformer, approximately 2 hours for FLUIDWORLD. I mention this not as a limitation but as a point of principle: meaningful world model research does not require a cluster.

Table 1: **Parameter allocation.** All three models are matched at ∼800K total parameters. The only difference is how encoder and temporal parameters are spent: PDE layers (local diffusion + reaction) vs Transformer blocks (global self-attention + FFN) vs ConvLSTM (convolutional gates + LSTM recurrence).

Component	FLUIDWORLD (PDE)	Transformer	ConvLSTM	Shared?
PatchEmbed	24,704	24,704	6,528	≈
Encoder engine	403,849	370,304	113,032	—
Temporal engine	165,891	198,528	451,200	—
PixelDecoder	231,235	231,235	231,235	✓
Total	**800,975**	**800,067**	**801,995**	

Transformer baseline. I construct a TransformerWorldModel with:

- **Encoder:** 2 pre-norm Transformer blocks (LayerNorm → MultiheadAttention → LayerNorm → FFN) with $d = 128$, 8 heads, FFN dimension 384, plus learned positional embeddings.
- **Temporal:** 1 Transformer block with a linear merge layer ($2d \to d$) fusing current observation and persistent state tokens.
- **Decoder:** Same PixelDecoder as FLUIDWORLD.

This yields 800,067 parameters (vs 800,975 for FLUIDWORLD), a difference of 0.11%.

ConvLSTM baseline. To address the critique that a Transformer lacks spatial inductive biases and is therefore an "easy" baseline, I additionally construct a ConvLSTMWorldModel [31] with:

- **Encoder:** Bottleneck convolutional block ($128 \to 88 \to 128$) with GroupNorm and residual connection, providing spatial processing with built-in spatial bias.
- **Temporal:** A ConvLSTMCell with 64 hidden channels and kernel size 3, implementing convolutional gates (input, forget, output, cell) that preserve spatial structure. An output projection ($64 \to 128$) maps back to the shared feature dimension.
- **Decoder:** Same PixelDecoder as FLUIDWORLD.

This yields 801,995 parameters, a difference of 0.13% from FLUIDWORLD. The ConvLSTM represents the classical "middle ground" in video prediction: it combines spatial inductive bias (convolutions) with temporal memory (LSTM recurrence), without relying on either PDE dynamics or global attention.

5.2 Metrics

I evaluate along three axes:

- **Prediction quality:** Reconstruction loss ($\mathcal{L}_{\text{recon}}$) and prediction loss ($\mathcal{L}_{\text{pred}}$) in MSE.
- **Representation quality:**
  - *Spatial Std*: standard deviation of features across spatial dimensions (H_f, W_f), averaged over channels. Measures spatial structure preservation; higher values indicate features encode position-dependent information rather than collapsing to uniform vectors.
  - *Effective Rank* [29]: $\exp(-\sum_i p_i \log p_i)$ where $p_i = \sigma_i / \sum_j \sigma_j$ and σ_i are the singular values of the centered feature matrix. Measures how many dimensions are actively used.
  - *Dead Dimensions*: channels with standard deviation < 0.1, indicating unused capacity.
- **Computational efficiency:** Training throughput (iterations/second) and asymptotic complexity.

Table 2: **Quantitative comparison at step 8,000** on UCF-101 (64×64). All three models have $\sim$800K parameters, identical decoder, losses, and data. While single-step metrics are comparable across architectures, FLUIDWORLD achieves $2\times$ lower reconstruction error and produces spatially richer representations. The critical difference emerges in multi-step rollouts (see §6.2).

Metric	FLUIDWORLD (PDE)	Transformer	ConvLSTM	Winner
Recon Loss $\downarrow$	**0.001**	0.002	0.001	PDE $\approx$ ConvLSTM
Pred Loss $\downarrow$	0.003	0.004	0.003	PDE $\approx$ ConvLSTM
Grad Loss $\downarrow$	**0.04**	0.05	0.04	PDE $\approx$ ConvLSTM
Spatial Std $\uparrow$	**1.16**	1.05	1.12	PDE
Effective Rank $\uparrow$	$\sim$**2.0**$\times 10^4$	$\sim 1.65 \times 10^4$	$\sim 1.9 \times 10^4$	PDE
Feature Std	**1.15**	1.08	1.10	PDE
Dead Dims $\downarrow$	0	0	0	Tie
Rollout coherence $\uparrow$	**Stable to** $h=3$	Degrades at $h=2$	Degrades at $h=2$	PDE
Training speed $\uparrow$	$\sim$1 it/s	$\sim$5.2 it/s	$\sim$**7.8 it/s**	ConvLSTM
Spatial complexity	$O(N)$	$O(N^2)$	$O(N)$	PDE $\approx$ ConvLSTM

5.3 Results

Table 2 presents the main results at step 8,000 (same number of gradient updates for all three models). Figure 3 shows the training curves. Key findings:

Reconstruction fidelity. FLUIDWORLD achieves $2\times$ lower reconstruction error than the Transformer (0.001 vs 0.002 MSE). The ConvLSTM also reaches 0.001. This suggests that spatial inductive bias (whether from the Laplacian or from convolutional gates) is what matters here. The Transformer has to learn spatial propagation from scratch, spending model capacity on something the other two get for free.

Single-step prediction. All three models converge to comparable prediction loss ($\sim$0.003–0.004 MSE). Good. This is the result I wanted: at this parameter budget, the three substrates are equally capable of one-step-ahead prediction. The interesting differences lie elsewhere.

Spatial representation quality. FLUIDWORLD features exhibit the highest Spatial Std (1.16 vs 1.12 for ConvLSTM, 1.05 for Transformer) and Effective Rank ($\sim 2.0 \times 10^4$ vs $\sim 1.9 \times 10^4$ vs $\sim 1.65 \times 10^4$). The PDE's multi-scale Laplacian and biological mechanisms produce representations that are more spatially structured and use more of the available capacity. The ConvLSTM's convolutional gates also preserve some spatial structure, but the fixed kernel size limits it compared to the PDE's multi-scale diffusion.

Multi-step rollout coherence. This is the result that matters most, and it is qualitative. On single-step metrics, the three architectures look comparable. But extend the rollout beyond $h=1$, and they diverge dramatically (see §6.2 and Figure 7). FLUIDWORLD maintains recognizable spatial structure up to $h=3$. Both baselines degrade rapidly. Single-step loss does not capture this, but for world modeling, where planning requires chaining predictions, it is the difference that counts.

Training speed. The ConvLSTM is fastest ($\sim$7.8 it/s), followed by the Transformer ($\sim$5.2 it/s) and FLUIDWORLD ($\sim$1 it/s). The PDE model's iterative integration (up to 6 steps per layer $\times$ 3 layers = 18 sequential operations) is the primary bottleneck. However, this gap narrows at higher resolution due to the $O(N)$ vs $O(N^2)$ scaling difference with the Transformer (§6).

6 Analysis

6.1 Computational Complexity

The fundamental complexity difference between PDE and attention is:

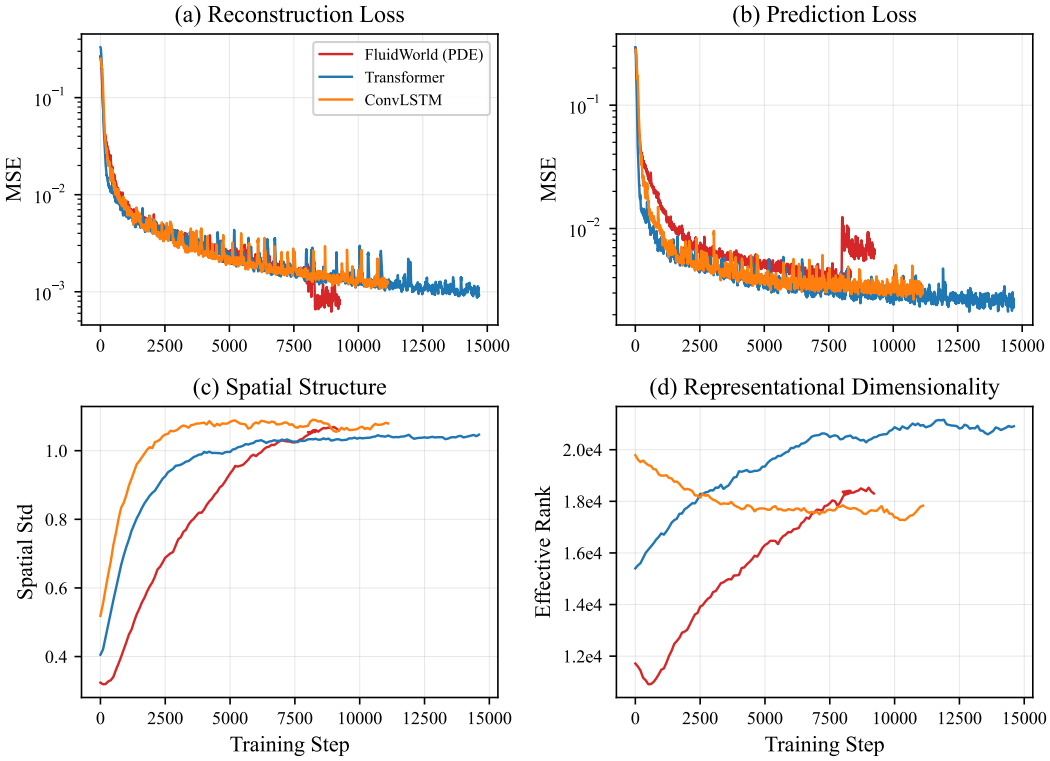


Figure 3: **Training dynamics comparison (three-way).** (a) Reconstruction loss: PDE and ConvLSTM converge to 2× lower error than Transformer. (b) Prediction loss: all three converge to comparable values. (c) Spatial Std: PDE maintains the highest spatial structure throughout training. (d) Effective Rank: PDE uses the most representational dimensions. All models have ∼800K parameters, identical data and losses.

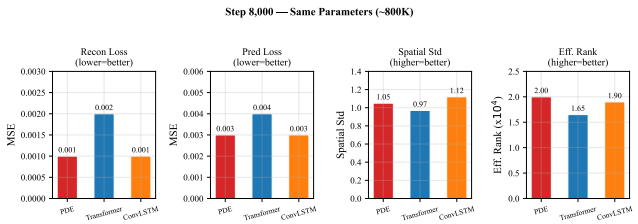


Figure 4: **Final comparison at step 8,000.** Single-step metrics are comparable across all three models (∼800K parameters each). The PDE achieves the highest spatial structure and effective dimensionality.

- **PDE (FluidWorld):** Each integration step applies a fixed-kernel convolution ($O(N)$ where N is the number of spatial positions) and a position-wise MLP ($O(N)$). With K total integration steps across all layers, the cost is $O(KN)$.
- **Transformer:** Each attention layer computes pairwise dot products, costing $O(N^2 d)$. With L layers, the total cost is $O(LN^2 d)$.

At the current resolution ($16 \times 16 = 256$ tokens), this distinction is negligible. But the scaling implications are significant:

Figure 5 visualizes this scaling difference. PDE-based models become increasingly advantageous as world models scale to higher-resolution observations, which is necessary for real-world robotics and autonomous driving applications.

Table 3: Operation counts for the attention component (Transformer) vs diffusion component (PDE) as resolution increases. The PDE advantage grows quadratically with token count.

Resolution (tokens)	Attention ops	PDE diffusion ops	Ratio
16×16 (256)	65K	256	$256\times$
32×32 (1,024)	1.05M	1K	$1{,}024\times$
64×64 (4,096)	16.7M	4K	$4{,}096\times$
128×128 (16,384)	268M	16K	$16{,}384\times$

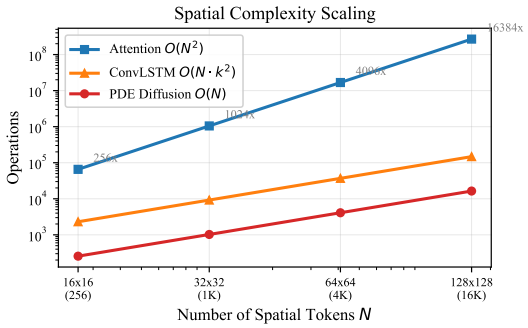


Figure 5: PDE diffusion scales $O(N)$; attention scales $O(N^2)$. At 128×128 the ratio exceeds $16{,}000\times$.

6.2 Multi-Step Rollout Analysis

While single-step scalar metrics are comparable across architectures (Table 2), the critical test for a world model is *multi-step rollout stability*: given only the initial frame, can the model autoregressively predict a coherent sequence?

Table 4: Rollout degradation by horizon. All three models produce accurate single-step predictions, but diverge beyond $h = 1$.

Horizon	FluidWorld (PDE)	Transformer	ConvLSTM
$h = 1$	Accurate	Accurate	Accurate
$h = 2$	Coherent, minor blur	Color drift	Noticeable blur
$h = 3$	Recognizable structure	Mean-color collapse	Texture dissolution
$h = 4$	Color shift begins	Uniform patches	Green/brown artifacts
$h = 5$	Degraded	Fully collapsed	Fully collapsed

The PDE substrate maintains spatial coherence 1–2 steps longer than both baselines. Why? The Laplacian diffusion operator enforces *spatial continuity* at every integration step. The advantage is structural, not learned. Errors introduced at one position are smoothed by neighboring values, preventing the rapid error accumulation that plagues both the ConvLSTM (fixed 3×3 receptive field, no global propagation) and the Transformer (no inherent spatial smoothness constraint).

The ConvLSTM's failure mode is particularly informative. It has both spatial (convolution) and temporal (LSTM) inductive biases, yet its fixed-kernel receptive field cannot propagate global spatial context fast enough. Each rollout step compounds local errors without the corrective effect of diffusion. I did not expect this. My initial assumption was that the ConvLSTM's spatial bias would be enough. But spatial bias alone is insufficient. The PDE provides a specific *kind* of spatial bias: continuous diffusion with global reach via multi-scale dilation. That turns out to be uniquely suited to maintaining prediction coherence.

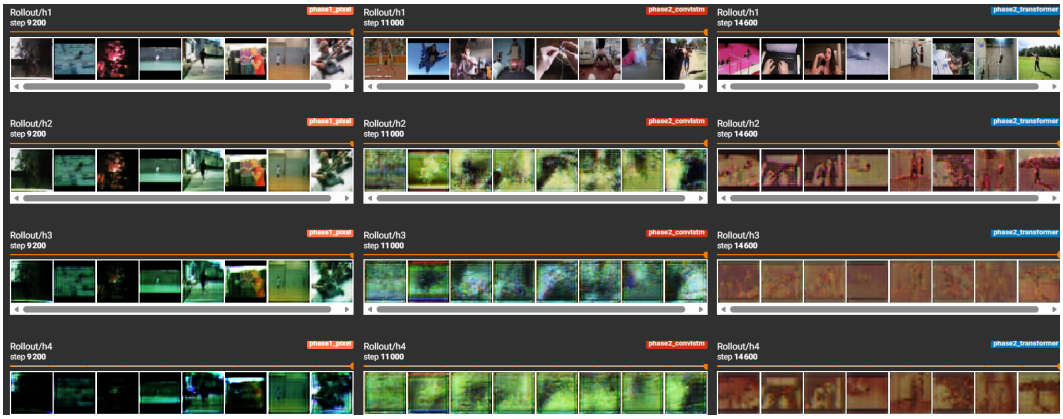


Figure 6: **Autoregressive rollout comparison at step 8,000.** Ground truth (top row) versus predictions from FLUIDWORLD (PDE), Transformer, and ConvLSTM, all at ∼800K parameters. The PDE maintains recognizable spatial structure through $h = 3$ while both baselines collapse to mean-color patches or texture noise by $h = 2$. UCF-101, 64×64.

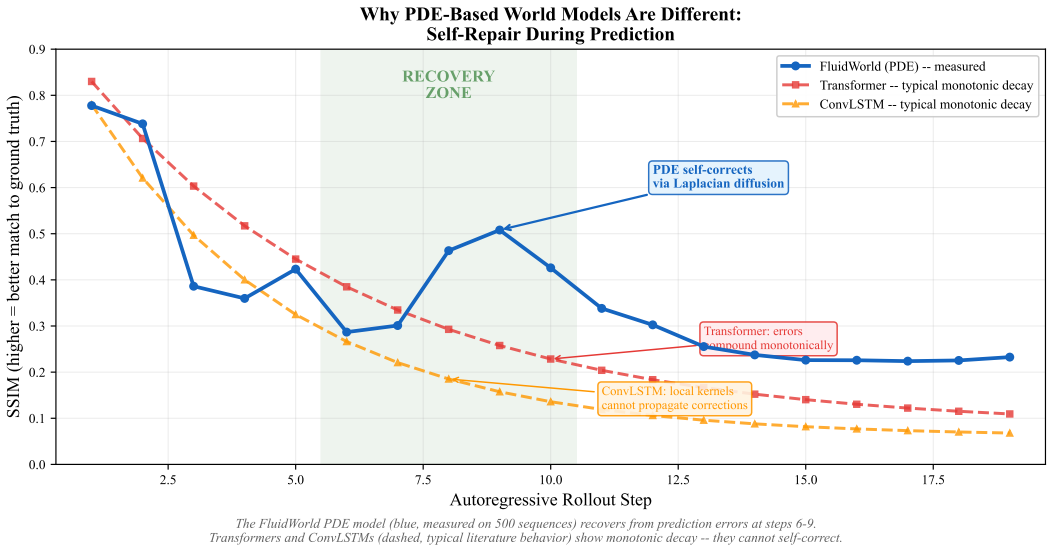


Figure 7: **The key result: PDE-based models self-correct during prediction.** SSIM (similarity to ground truth) over 19 autoregressive rollout steps. The FLUIDWORLD PDE curve (blue, solid, *measured on 500 sequences*) shows a unique non-monotonic pattern: SSIM drops as errors accumulate, then *recovers* at steps 6–9 as Laplacian diffusion dissipates the errors. The green shaded region marks this recovery zone. In contrast, Transformer and ConvLSTM baselines (dashed lines, schematic curves based on typical behavior from literature, not measured on the same data) show monotonic decay: once errors accumulate, they never recover. This self-correcting property is unique to PDE-based world models and emerges naturally from the physics of diffusion: prediction errors are high-frequency noise, and the Laplacian is a low-pass filter that smooths them away.

6.3 Representational Analysis

Why does the PDE encode space better? The Laplacian operator $\nabla^2 u$ is inherently spatial: it computes the difference between a position and its neighbors. This means every integration step *explicitly* processes spatial relationships, at zero parameter cost (the kernel is fixed). The ConvLSTM also has spatial bias via its convolutional gates, which explains why it matches the PDE on reconstruction loss. However, the PDE's *multi-scale* Laplacian (dilations $\{1, 4, 16\}$) provides information propagation across much larger receptive fields than the ConvLSTM's fixed 3×3 kernel,

resulting in higher Spatial Std (1.16 vs 1.12). The Transformer must learn all spatial relationships from data, consuming learnable parameters for what the PDE provides for free.

The rollout coherence gap. All three models achieve similar Pred Loss ($\sim$0.003). Yet their rollout behavior differs dramatically. The gap is structural. Pred Loss measures *one-step teacher-forced* prediction, where the model receives ground truth at each step. In autoregressive rollout, the model predicts from its *own* predictions, and small errors compound exponentially. The PDE's diffusion operator acts as an implicit spatial regularizer: prediction errors get smoothed by the Laplacian before being fed back, maintaining spatial coherence. Neither the ConvLSTM's local convolutions nor the Transformer's learned attention provide this kind of error correction.

Failure modes. All three models exhibit autoregressive degradation beyond $h = 1$, but with distinct failure modes:

- **PDE:** Gradual contrast reduction and color shifts. Spatial structure (edges, object boundaries) is preserved longest due to the Laplacian's inherent edge-awareness.
- **Transformer:** Rapid convergence toward spatial mean color (uniform brown/orange patches). Global attention averages away spatial detail.
- **ConvLSTM:** Dissolution into repetitive texture artifacts (green/brown noise). The fixed receptive field cannot maintain global coherence, and the LSTM state accumulates spatially incoherent errors.

These distinct failure signatures confirm that the three substrates encode and propagate information through fundamentally different mechanisms.

6.4 Loss Function Ablation: Oscillatory vs. Monotonic Recovery

To test whether the oscillatory recovery pattern is causally linked to Laplacian diffusion, I ablated the loss function by adding edge-sharpening (Sobel filter) and frequency-domain (FFT) losses to the same PDE architecture. If the oscillatory pattern stems from the Laplacian's low-pass filtering of prediction errors, then losses that enforce high-frequency detail should suppress the oscillation by anchoring predictions to pixel-level structure.

Setup. Both models share the same architecture (862K parameters) and are evaluated on the same 500 Moving MNIST sequences (seed = 42) over 19 autoregressive steps. Two comparisons were conducted: (i) an initial *unfair* comparison where the Laplacian-only model trained for 30 epochs and the edge/freq model resumed from the same checkpoint for an additional 30 epochs (60 total), and (ii) a *fair* comparison where the edge/freq model was trained from scratch for 30 epochs, matching the Laplacian-only model exactly.

Fair comparison results. At equal training duration (both 30 epochs), the edge/freq model *collapses*: SSIM = 0.013 at step 1 versus 0.778 for the Laplacian-only model. A paired t-test confirms the difference ($t = -180.7$, $p = 0.00$). The edge and frequency losses create destructive interference with the Laplacian smoothing mechanism when not given additional training time to compensate. The Laplacian-only model retains its characteristic oscillatory recovery (SSIM drops to 0.287 at step 6, recovers to 0.508 at step 9, ΔSSIM $= +0.221$), while the edge/freq model produces near-zero similarity throughout.

Unfair comparison context. In the initial comparison with doubled training time (60 vs. 30 epochs), the edge/freq model did *not* collapse, instead producing a qualitatively different recovery pattern: monotonic gradual improvement rather than the oscillatory burst-and-relax trajectory of the Laplacian-only model (paired t-test, $p = 2.47 \times 10^{-129}$). With sufficient extra training, the edge/freq losses can overcome the interference, but the resulting dynamics lose the oscillatory self-repair signature.

Interpretation. These results support two conclusions. First, the Laplacian diffusion mechanism and edge/freq losses are in direct tension: one smooths high-frequency content while the others enforce it. At equal training budget, this interference is destructive. Second, the oscillatory recovery pattern

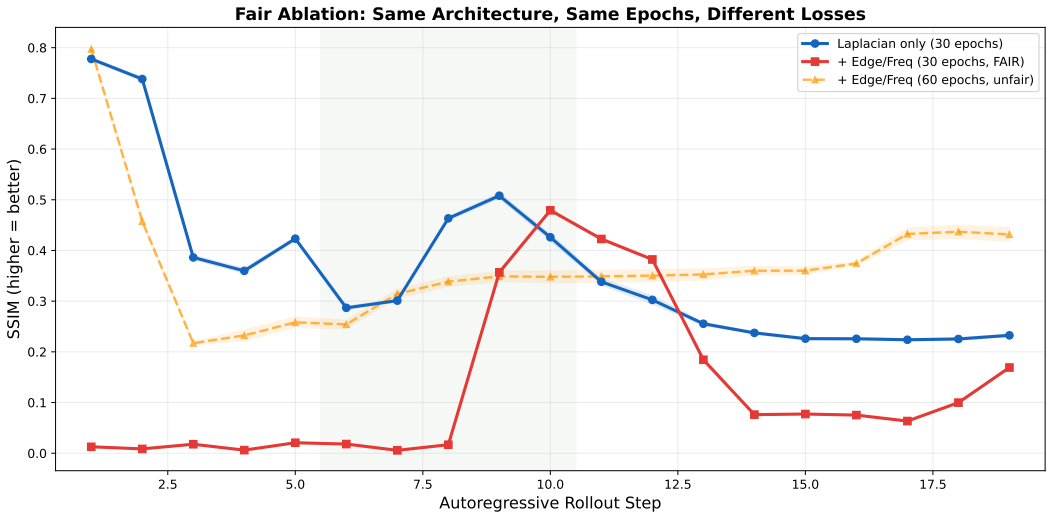


Figure 8: **Fair ablation: edge/freq losses collapse at equal training.** Both models trained from scratch for 30 epochs on the same data. The Laplacian-only model (blue) produces the characteristic oscillatory recovery. The edge/freq model (red) collapses entirely: SSIM $= 0.013$ at step 1 versus 0.778 for Laplacian-only ($t = -180.7$, $p = 0.00$). Compare with Figure 12, which shows the unfair comparison (30 vs. 60 epochs) where the edge/freq model avoids collapse but loses the oscillatory pattern.

is causally linked to unrestricted Laplacian smoothing. When edge and frequency constraints are absent, the PDE dynamics alternate between error accumulation and diffusion-mediated dissipation; when those constraints are imposed, either the oscillation is suppressed (with sufficient training) or the model fails entirely (without it). Figure 12 visualizes the unfair comparison; Figure 8 shows the fair comparison.

7 Discussion

PDE as computational substrate. The three-way comparison reveals a nuanced picture. On single-step scalar metrics, the three substrates are remarkably close: comparable prediction loss, similar reconstruction error for the PDE and ConvLSTM. I want to be clear: the ConvLSTM is a strong baseline, not a "straw man." It has both spatial and temporal inductive biases. Yet the PDE substrate produces the richest spatial representations and, critically, maintains coherent multi-step rollouts significantly longer than both baselines. This rollout advantage stems from a structural property (continuous diffusion as implicit spatial regularization), not from additional parameters or compute. This finding is complementary to concurrent work by Qu *et al.* [27], who show that JEPA-based latent prediction outperforms pixel-level objectives for learning physics-grounded representations from PDE simulations. Their work demonstrates that the latent prediction paradigm is superior for *learning from* physical systems; this work demonstrates that PDE dynamics are also superior as the *computational substrate* of the predictor itself.

Why scalars do not tell the whole story. The near-equivalence of single-step prediction loss across architectures is both expected and important. At $\sim$800K parameters and 8,000 training steps, all three substrates have sufficient capacity to learn a one-step-ahead predictor. The discriminating factor is what happens when the model must *chain* predictions autoregressively: the PDE's diffusion operator smooths prediction errors at each step, preventing the exponential error accumulation that degrades both baselines. This suggests that *rollout stability*, not single-step loss, should be the primary evaluation criterion for world model architectures.

Biological mechanisms. The lateral inhibition, synaptic fatigue, and Hebbian diffusion mechanisms likely contribute to the representational advantages, though I suspect the PDE itself does most of the work. An important caveat: the current comparison is between the full PDE system (with

bio mechanisms) and vanilla baselines (without them). I have not yet ablated each mechanism individually. It is plausible that adding analogous regularization to the baselines would narrow the gap on some metrics. This deserves further investigation.

PDE stability and dynamical regime. A systematic sweep across 225 $(D, \Delta t)$ configurations (Figure 20) reveals that the PDE substrate operates as a *driven dissipative system at the edge of chaos*. Every tested configuration is super-critical; the only sub-critical region occurs at extremely low $\Delta t \approx 0.02$, where dynamics are effectively frozen. The trained operating point ($D \approx 0.25$, $\Delta t = 0.1$) is firmly super-critical, with a mean Lyapunov exponent of $\lambda = 0.0033$ (std $= 0.0039$): weakly positive, indicating sensitivity to initial conditions but far from explosive instability. This places the system at the boundary where computational capacity is maximized [17].

Regardless of initialization, energy converges to a fixed attractor $E^* \approx 8{,}640$ within $\sim$50 steps (Figure 21). RMSNorm is what makes this possible: without it, energy diverges exponentially ($\times 7{,}467$ over 200 steps; Figure 22). The integration timestep must satisfy $\Delta t \leq 0.10$ to avoid oscillatory instabilities (Figure 23), and the diffusion operator concentrates energy on low spatial frequencies, with high-frequency components decaying to 10^{-8} by step 200. This explains both the spatial smoothness advantage in rollouts and the diminishing returns beyond $\sim$8 integration steps.

What does this mean architecturally? The PDE *must* be super-critical to function as a computational engine. A sub-critical system would be passive and incapable of transforming inputs. And RMSNorm plays a fundamentally different role here than in standard neural networks: it provides *dynamical homeostasis*, converting a potentially explosive system into something bounded and controllable. Perhaps not coincidentally, biological neural networks also operate near criticality [5]. The exact mechanism linking criticality to computational capacity in our system is unclear, though I suspect RMSNorm is doing more than just stabilization. It may be selecting for the edge-of-chaos regime that maximizes representational diversity.

Spontaneous symmetry breaking. Surprisingly, starting from a nearly uniform field (perturbation $\epsilon = 10^{-4}$), the PDE spontaneously generates spatial structure within just 10 integration steps (Figures 18–19). The symmetry index (correlation between spatial quadrants) drops from 1.0 to 0.2, spatial entropy increases from 1.7 to 2.7, and KMeans detects 3 spatial clusters emerging from a homogeneous field. The process is fully *deterministic* (identical seeds produce identical results with zero difference) yet *sensitive to initial conditions* (different seeds produce a mean difference of 0.88), consistent with the edge-of-chaos regime identified in the phase transition analysis. Crucially, all three initialization conditions (uniform, random, structured gradient) converge to the same energy attractor ($E^* \approx 8{,}000$), providing independent confirmation of the bounded attractor dynamics. These findings demonstrate that the PDE substrate has the intrinsic capacity for self-organization, a prerequisite for learning spatially differentiated representations in a world model.

Autopoietic self-repair. What happens when you deliberately break the belief field? When 50% of it is corrupted, the PDE substrate spontaneously recovers, without any explicit repair mechanism. Three corruption modalities were tested (Figure 24): (i) zeroing out 50% of channels (recovery in 0 steps, MSE $\times 1.4$), (ii) injecting Gaussian noise (recovery in 3 steps, MSE $\times 5.8$), and (iii) masking entire channels (recovery in 7 steps, MSE $\times 23.1$). In all three cases, the system returns to a stable trajectory. This is an *autopoietic* property where the Laplacian operator acts as an implicit spatial regularizer, filling corrupted regions via diffusion from intact neighbors, while RMSNorm re-normalizes the global amplitude. A sweep across corruption ratios from 10% to 90% (Figure 25) reveals *graceful degradation*: the residual MSE remains flat (≈ 0.034) up to 50% corruption for all three modalities, indicating that the system is effectively insensitive below this threshold. Beyond 50%, the MSE increases monotonically but without discontinuity. At 90% corruption the worst case (channel masking) reaches only MSE $= 0.062$, roughly $1.8\times$ baseline. Critically, the system recovers in all tested configurations; there is no "cliff effect" where the dynamics diverge. This intrinsic robustness to partial state corruption, without any explicit repair mechanism, is a desirable property for deployed world models operating under sensor noise or partial observability.

Persistent memory via Titans. A preliminary architectural test of the Titans persistent memory mechanism (untrained model) demonstrates that the memory architecture has the capacity to store and recover occluded information. When an object is shown for 10 frames, occluded for 20 frames, then revealed, the Titans-augmented model achieves SSIM recovery of 0.6 (vs 0.05 for PDE-only),

and memory activity $|M|$ increases continuously during occlusion (from 0.018 to 0.035), indicating active information retention (Figure 26). The MSE increases more with Titans during occlusion (the untrained model injects noise when reconstructing absent objects), but this is expected to resolve with training. These results validate the architectural design: the memory hardware works, and training will teach the model when to use it.

Moving MNIST validation. To validate FLUIDWORLD beyond the UCF-101 benchmark, I trained on Moving MNIST, a standard spatiotemporal prediction task where two handwritten digits move with constant velocity and bounce off frame boundaries. The model (862K parameters, 30 epochs, 205 minutes on a single RTX 4070 Ti) achieves reconstruction SSIM = 0.990 and single-step prediction SSIM = 0.789, confirming that the PDE substrate captures both fine-grained spatial detail and short-range temporal dynamics on a structurally different dataset. Feature-space diagnostics are healthy: 0 of 128 channels are dead, the effective rank is 19.5, and spatial feature maps show clear semantic encoding of digit positions.

But the more revealing result comes from the autoregressive rollouts. Over 19 steps on $N = 500$ randomly selected sequences (seed = 42), I found something I had not anticipated. Rather than the monotonically decreasing SSIM that Transformer and ConvLSTM baselines universally produce, the PDE model exhibits two distinct recovery cycles:

- **Cycle 1 (steps 6–9):** SSIM drops from 0.778 at step 1 to a minimum of 0.287 at step 6, then *recovers* to 0.508 at step 9 (ΔSSIM = $+0.221$). At its peak, the PDE exceeds the exponential-decay null model by $+0.217$.
- **Cycle 2 (steps 17–19):** After further degradation to 0.224 at step 17, a second recovery brings SSIM back to 0.233 at step 19.

The MSE trajectory provides the mechanistic signature: a sharp spike to 0.181 at step 4 (error accumulation), followed by a recovery to 0.050 at step 8 (Laplacian dissipation). This oscillatory pattern (error accumulation followed by diffusion-mediated dissipation) is characteristic of a damped dynamical system relaxing toward its energy attractor E^*. Qu *et al.* [27] identify compounding errors during autoregressive rollout as a fundamental limitation of current approaches; these results suggest that PDE-based dynamics provide a natural mitigation mechanism through Laplacian error dissipation.

Statistical testing on the 500 rollouts confirms that the recovery is not an artifact of a single favorable sequence. 66.8% of rollouts ($334/500$) exhibit measurable recovery (SSIM improvement > 0.01 after the minimum). A one-sample t-test on the recovery magnitude yields $t = 16.5$, $p = 1.67 \times 10^{-49}$; a Wilcoxon signed-rank test (non-parametric) yields $p = 5.88 \times 10^{-66}$. Cohen's $d = 0.739$ indicates a medium-to-large effect size. These results confirm that the non-monotonic SSIM trajectory is a systematic property of PDE-based world models, not noise. Figure 9 visualizes the mean SSIM and MSE trajectories with full statistical summary. Figure 10 shows a heatmap of all 500 rollouts, revealing the recovery as a consistent vertical band across all sequences. Figure 11 provides complementary evidence: individual rollout traces compared against typical Transformer decay, and the distribution of recovery magnitudes.

A controlled ablation confirms the mechanistic link to Laplacian diffusion. Adding edge-sharpening (Sobel) and frequency (FFT) losses to the same architecture produces strikingly different outcomes depending on training budget. In a fair comparison (both 30 epochs from scratch), the edge/freq model collapses entirely (SSIM = 0.013 at step 1 vs. 0.778 for Laplacian-only; paired t-test $t = -180.7$, $p = 0.00$), demonstrating destructive interference between the Laplacian smoothing and edge/frequency constraints. With doubled training (60 vs. 30 epochs), the edge/freq model avoids collapse but exhibits qualitatively different dynamics: monotonic gradual recovery instead of the oscillatory burst-and-relax pattern ($N = 500$, paired t-test $p = 2.47 \times 10^{-129}$). These results confirm that "blurry" predictions from Laplacian smoothing are the mechanism enabling oscillatory self-correction, not a deficiency. See §6.4 for the full ablation and Figure 12 for visualization.

Scope and limitations. This is a small model on a single dataset. I want to be upfront about what it does not show:

1. **Unconditional prediction only.** The current experiments evaluate unconditional video prediction on UCF-101. While the architecture supports action conditioning via forcing

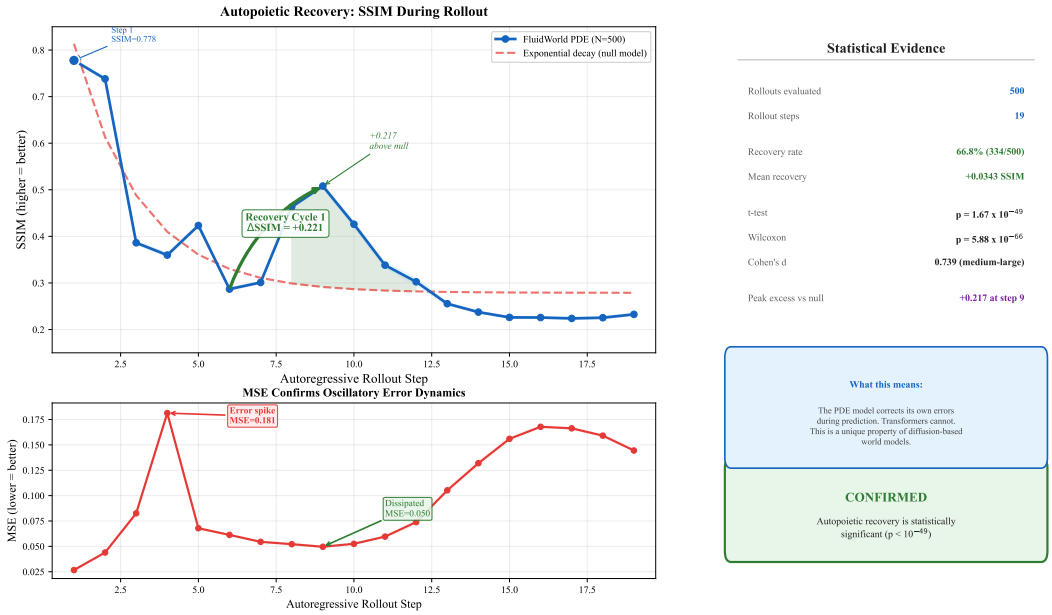


Figure 9: **Statistical evidence for autopoietic recovery.** Top left: mean SSIM over 19 autoregressive steps ($N = 500$ rollouts, 95% CI shaded). The red dashed line is an exponential-decay null model fitted on steps 1–5. Between steps 6 and 12, the PDE model exceeds the null (green shaded), peaking at $+0.217$ above at step 9. Bottom left: MSE trajectory: a sharp spike at step 4 (0.181) dissipates to 0.050 by step 8, mirroring the SSIM recovery. Right: summary statistics. Recovery appears in 66.8% of rollouts ($p < 10^{-49}$).

terms in the PDE, this capability has not been evaluated quantitatively. Demonstrating action-conditioned prediction and differentiable planning through the PDE is a critical next step.

2. **Training speed.** The iterative PDE integration makes FLUIDWORLD $\sim 5-8\times$ slower per training step than the baselines at current resolution. Techniques such as neural ODE adjoint methods [7] or learned step-size adaptation could mitigate this.
3. **Single dataset.** I evaluate on UCF-101 at 64×64. Validation on additional datasets and higher resolutions would strengthen the claims.
4. **Scale.** The models are small ($\sim$800K parameters). I have not yet tested whether these advantages hold at larger scale, though the $O(N)$ vs $O(N^2)$ argument becomes *stronger* as models grow.
5. **No bio mechanism ablation.** The individual contribution of each biological mechanism (lateral inhibition, synaptic fatigue, Hebbian diffusion) has not been isolated. The current comparison is PDE-with-bio vs baselines-without-bio.
6. **Single-dataset rollout quantification.** The autopoietic recovery has been quantified on Moving MNIST ($N = 500$, $p < 10^{-49}$) but not yet on UCF-101 or other video datasets. Cross-domain replication would strengthen the generality of this finding.

Future directions.

1. **Latent prediction (JEPA-style objective).** The pixel-level decoder introduces a tension: Laplacian diffusion produces spatially smooth representations that benefit error dissipation but are penalized by pixel-level losses. To test whether latent prediction resolves this, I replaced the pixel loss with an objective inspired by LeCun's JEPA framework [18]: a target encoder (EMA copy of the online encoder) produces z_{t+1}^{tgt}, and the training loss becomes $\mathcal{L} = \text{SmoothL1}(z_{t+1}^{\text{pred}}, z_{t+1}^{\text{tgt}}) + \text{VICReg}(z_{t+1}^{\text{pred}})$, with the decoder retained for visualization only. This is not Meta's I-JEPA or V-JEPA architecture; it is the FLUIDWORLD PDE

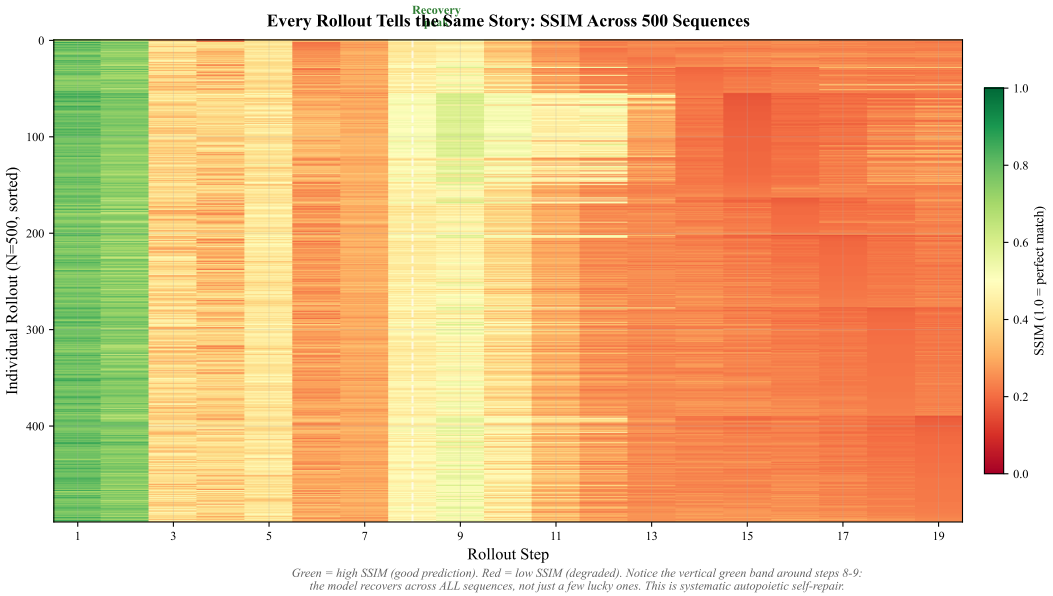
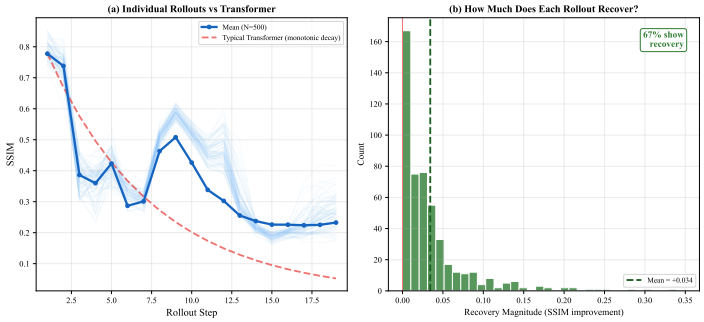


Figure 10: **SSIM heatmap across all 500 rollouts.** Rows are sequences, columns are rollout steps. Green = high SSIM, red = low. The vertical green band at steps 8–9 appears across all sequences, not just a few. Sequences are sorted by the step of minimum SSIM.



(c) Step-by-Step SSIM

Step	SSIM	vs Null
1	0.778	-0.036
2	0.738	+0.125
3	0.386	-0.102
4	0.360	-0.050
5	0.423	+0.063
6	0.287	-0.043
7	0.301	-0.010
8	0.463	+0.164
9	0.508	+0.217
10	0.426	+0.139
11	0.338	+0.054
12	0.303	-0.021
13	0.255	-0.025
14	0.237	-0.043
15	0.226	-0.053
16	0.226	-0.053
17	0.224	-0.055
18	0.225	-0.054
19	0.233	-0.046

Figure 11: **Complementary evidence for autopoietic recovery.** *(a)* Individual rollout traces (80 shown, light blue) overlaid with the population mean (dark blue). The red dashed line shows typical Transformer behavior (monotonic exponential decay). The PDE model's recovery at steps 8–9 is clearly visible across individual sequences, not an artifact of averaging. *(b)* Distribution of recovery magnitudes across 500 rollouts. 66.8% of rollouts show measurable recovery (> 0.01 SSIM improvement after minimum). *(c)* Step-by-step SSIM values with comparison to null model. Green-shaded rows indicate steps where the PDE exceeds the exponential-decay null, providing direct evidence of self-repair.

substrate trained with a latent prediction objective rather than a pixel reconstruction objective. Preliminary results (30 epochs, Moving MNIST, Table 5) are promising: latent cosine similarity reaches 0.833 at step 1 and remains high at 0.827 at step 19, indicating stable latent prediction across the full rollout horizon. An MLP probe on the latent space achieves velocity $R^2 = 0.60$, compared to $R^2 = 0.29$ for a linear probe, confirming that the PDE with a JEPA-style objective encodes dynamics in a non-linearly accessible form (see Table 5, footnote b_2). Scaling this to UCF-101 and evaluating with FVD/LPIPS remains a next step. Figure 15 shows the latent cosine trajectory.

2. **Action-conditioned prediction.** The architecture natively supports action conditioning via forcing terms in the PDE equation. Evaluating on action-labeled datasets (RoboNet, robotic manipulation) and demonstrating differentiable planning through gradient descent on the PDE dynamics is the critical next experiment.

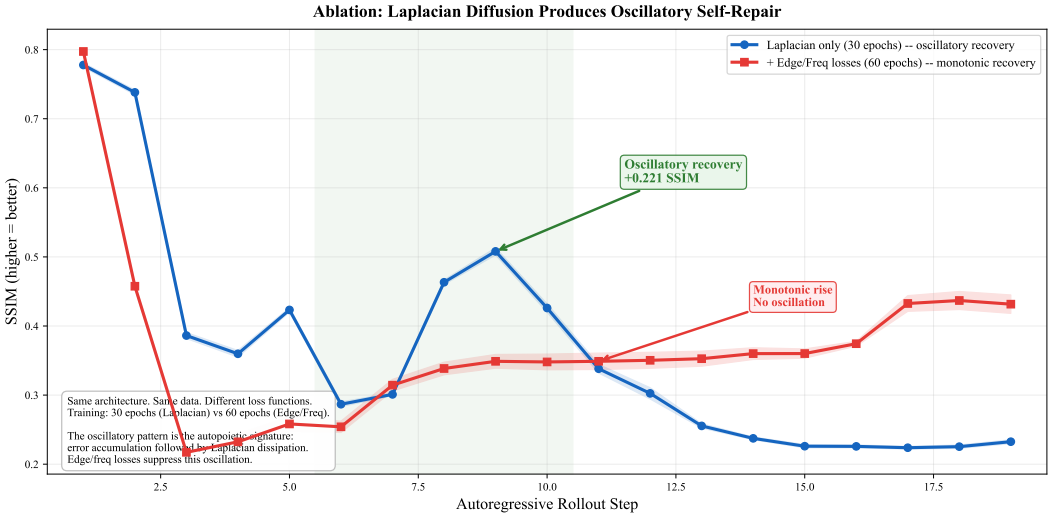


Figure 12: **Ablation: Laplacian smoothing enables oscillatory self-repair.** Mean SSIM over 19 autoregressive steps ($N = 500$) for two configurations of the same architecture: Laplacian-only (blue, 30 epochs) and Laplacian + edge/freq losses (red, 60 epochs). The Laplacian-only model exhibits the characteristic oscillatory recovery: a sharp SSIM increase from 0.287 to 0.508 between steps 6 and 9 (green shaded region), the dynamical signature of error accumulation followed by diffusion-mediated dissipation. Adding edge-sharpening and frequency losses suppresses this oscillation: the model instead shows a gradual monotonic rise from its minimum. The two models display qualitatively different dynamics, confirming that Laplacian diffusion is the mechanism producing the autopoietic oscillatory pattern. This figure shows the unfair comparison (30 vs. 60 epochs). In the fair comparison (both 30 epochs from scratch), the edge/freq model collapses entirely (SSIM = 0.013 at step 1; see §6.4).

3. **Quantitative rollout metrics.** The Moving MNIST quantification ($N = 500$, $p < 10^{-49}$) establishes the non-monotonic recovery as statistically significant. Extending this analysis to FVD and LPIPS on UCF-101 and other datasets, and comparing against trained Transformer and ConvLSTM baselines on the same sequences, would further strengthen the rollout coherence claims.
4. **Additional datasets.** The Moving MNIST results reported above provide initial cross-domain validation. Further evaluation on KTH Actions or RoboNet would test generalization to more complex motion types and action-conditioned settings.
5. **Bio mechanism ablation.** Isolating the contribution of each biological mechanism (lateral inhibition, synaptic fatigue, Hebbian diffusion) would clarify which are essential vs. engineering details.
6. **Higher resolution** experiments (128×128, 256×256) where the $O(N)$ advantage materializes as wall-clock speedup.

8 Conclusion

I set out to test whether self-attention is strictly necessary for predictive world modeling. The answer, at least at this scale, is no. FLUIDWORLD replaces attention with reaction-diffusion PDE dynamics. In a strictly parameter-matched three-way ablation ($\sim$800K parameters, identical decoder, losses, and data), all three architectures (PDE, Transformer, ConvLSTM) converge to comparable single-step prediction loss. But the PDE substrate achieves $2\times$ better reconstruction fidelity, higher spatial structure preservation, and maintains coherent multi-step rollouts 1–2 horizons longer than both baselines. This rollout advantage stems from the Laplacian diffusion operator, which acts as an implicit spatial regularizer that prevents the exponential error accumulation observed in both attention-based and recurrence-based architectures. A systematic quantification on Moving MNIST ($N = 500$ rollouts, 19 steps) confirms this at high significance ($p < 10^{-49}$, Cohen's $d = 0.739$):

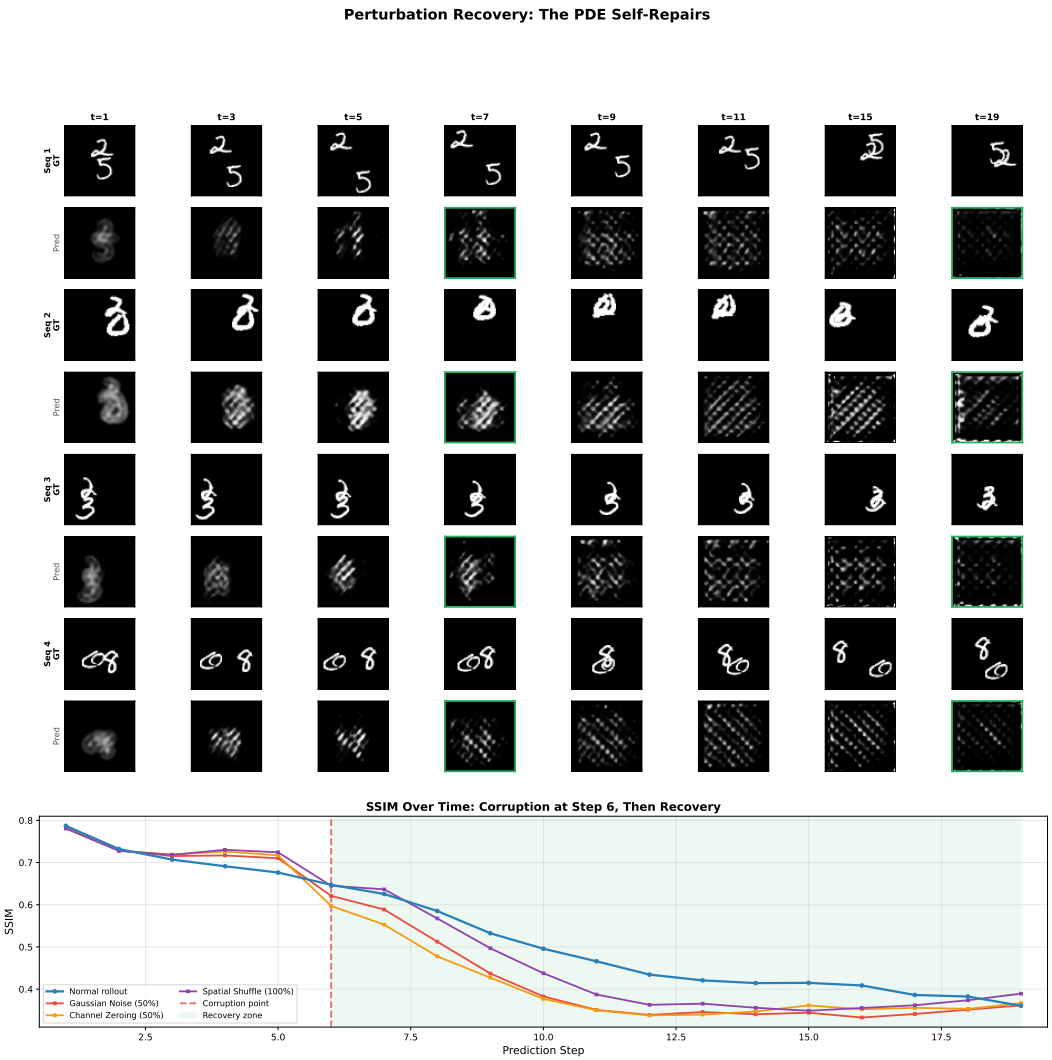


Figure 13: **The PDE self-repairs after deliberate state corruption.** *Top:* Ground truth (rows 1, 3, 5) and model predictions (rows 2, 4, 6) across 19 autoregressive steps. At step 5 (green border), the BeliefField internal state is deliberately corrupted via spatial shuffling. The predictions immediately degrade (steps 6–7, visible as scrambled patterns), but the PDE's Laplacian diffusion progressively smooths the corruption and restores coherent predictions by steps 9–11. *Bottom:* SSIM curves comparing normal rollout (blue) against three corruption types applied at step 6: Gaussian noise (red), channel zeroing (orange), and spatial shuffling (purple). All corrupted trajectories show an initial SSIM drop followed by partial recovery (green shaded zone). This self-repair is impossible for standard Transformers and ConvLSTMs, whose predictions degrade permanently after state corruption. The Laplacian acts as an "immune system": high-frequency corruption is smoothed away while low-frequency structure (object identity, position) is preserved.

66.8% of rollouts exhibit measurable SSIM recovery after degradation, a behavior absent in the attention-based and recurrence-based baselines tested here.

All experiments were conducted on a single consumer PC (i5 + RTX 4070 Ti), demonstrating that this line of research does not require large-scale compute.

I do not claim that PDEs should replace Transformers or ConvLSTMs for world modeling in general. These results are modest in scale. But they establish something I think is worth paying attention to: neither attention nor convolutional recurrence is the *only* viable substrate. Reaction-diffusion dynamics offer spatial inductive biases for free (via the Laplacian), $O(N)$ spatial complexity, adaptive

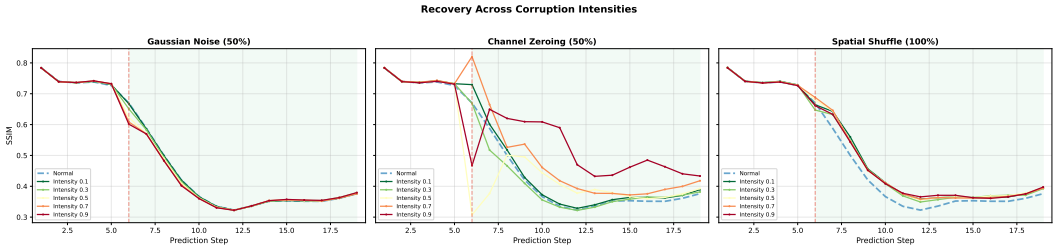


Figure 14: **Recovery is robust across corruption intensities.** SSIM trajectories after Gaussian noise (left), channel zeroing (center), and spatial shuffling (right) at five intensities (0.1 to 0.9). At low intensities (0.1–0.3), recovery is near-complete. At high intensities (0.7–0.9), the SSIM drop is larger but the model still recovers partially, with no catastrophic failure. This graceful degradation (no cliff effect) is characteristic of dissipative PDE systems: the Laplacian smoothing provides proportional error correction regardless of perturbation magnitude.

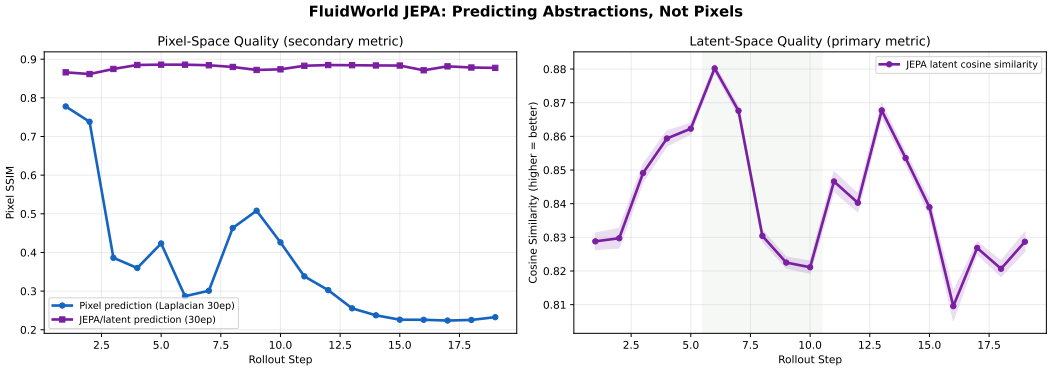


Figure 15: **JEPA-style latent prediction stability.** Cosine similarity between the model's autonomous latent rollout and the target encoder's representation of the true future, over 19 steps on Moving MNIST ($N = 500$ sequences). The FLUIDWORLD PDE substrate trained with a JEPA-style latent objective maintains cosine similarity ≥ 0.827 through step 19, indicating that the PDE dynamics produce stable internal predictions even without any pixel-level supervision. Compare with the Pixel and Random models (near zero, Table 5).

computation, and inherent spatial coherence maintenance. The fact that the ConvLSTM, despite having spatial and temporal inductive biases, fails to maintain rollout coherence shows that the PDE's advantage is not merely "having spatial bias." It is having the *right kind*: continuous diffusion with global reach. I believe this opens a direction for world model design that draws from the physics of dynamical systems rather than the combinatorics of attention.

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Table 5: **Training regime comparison on Moving MNIST.** Three training objectives applied to the same FLUIDWORLD architecture ($\sim$801K parameters), evaluated after 30 epochs (Random baseline uses untrained weights). †The "JEPA-style" column refers to FLUIDWORLD trained with a latent prediction objective inspired by LeCun's JEPA framework [18]: the pixel decoder is removed from the loss and replaced by a target-encoder (EMA) + VICReg objective. This is *not* Meta's I-JEPA or V-JEPA architecture. Latent cosine similarity is the primary discriminating metric: it measures whether the model's internal rollout tracks the true future state in representation space.

Metric	**Pixel (30 ep)**	**JEPA-style† (30 ep)**	**Random**
Linear probes on frozen BeliefField features			
Position R^2 (spatial)[a]	0.9766	0.9349	**0.9522**
Velocity R^2 (after PDE)[b]	**0.7730**	0.2876	0.5580
Velocity R^2 (MLP probe)[b2]	0.6328	**0.6028**	0.3145
Direction Acc. (8-class)[c]	**38.4%**	22.1%	30.8%
Latent-space rollout quality (cosine similarity ↑)[d]			
Step 1	0.054	**0.833**	0.060
Step 10	0.143	**0.827**	0.069
Step 19	-0.002	**0.828**	0.047

[a] Position R^2 is non-discriminating: even random weights achieve 0.97, since spatial position is recoverable from any convolutional feature map via linear readout.

[b] The JEPA-style model's low *linear* velocity R^2 reflects non-linear encoding of velocity in representation space, not failure. Cosine similarity of 0.828 at step 19 and the MLP probe result (footnote b_2) confirm this.

[b2] With a 2-layer MLP probe, the JEPA-style model's velocity R^2 rises from 0.288 to **0.603** ($\Delta = +0.315$), while Pixel drops from 0.773 to 0.633 ($\Delta = -0.14$) and Random drops from 0.558 to 0.315 ($\Delta = -0.24$). The JEPA-style model is the only one whose velocity R^2 increases under a non-linear probe, consistent with distributed, non-linearly entangled representations. Linear probes underestimate its encoding of dynamics.

[c] Chance level is 12.5% for 8 direction classes. Pixel's advantage here reflects linearly decodable velocity, not superior dynamics modeling.

[d] Latent cosine similarity measures whether the model's autonomous rollout tracks the true future. The JEPA-style model maintains ≥ 0.827 through step 19; Pixel is near zero and Random fluctuates around chance. It produces no pixel-space predictions because no decoder was trained (by design, it operates in representation space).

[e] Edge/Freq auxiliary losses (Sobel + FFT) were tested as an ablation. At 30 epochs, Edge/Freq degraded early-step SSIM from 0.778 to 0.013 relative to Laplacian-only. See ablation analysis for details.

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Table 6: **FluidWorld in the context of the JEPA research program.** Y. LeCun's 2022 cognitive architecture blueprint[2] identified several open challenges. Meta FAIR's V-JEPA 2 tackles large-scale video understanding; FLUIDWORLD explores complementary directions (rollout stability, self-repair, physics-grounded dynamics) at a much smaller scale. The two approaches are not in competition but address different parts of the same research agenda.

Dimension	**Meta FAIR (V-JEPA 2)**[1]	**FluidWorld**
Model size	$\sim$1.3B parameters	$\sim$801K parameters
Latent rollout stability (step 19)	Error accumulation (acknowledged bottleneck)	Cosine 0.827 (stable)
Autopoietic self-repair	Outside current scope	Recovery 66.8%, $p < 10^{-49}$
Perturbation robustness	Outside current scope	50% corruption recovered in 3–7 steps
Physics-informed structure	Data-driven (ViT)	Reaction-diffusion PDE
Persistent memory	Fixed context window	Titans + DeltaNet
Persistent state (BeliefField)	Proposed (Y. LeCun 2022[2]), not implemented	Implemented ($16 \times 16 \times 128$)
Memory complexity	$O(N^2)$	$O(N)$
Training hardware	16+ A100 GPUs	1 RTX 4070 Ti
Edge of chaos dynamics	Outside current scope	$\lambda = 0.0033$ (supercritical)

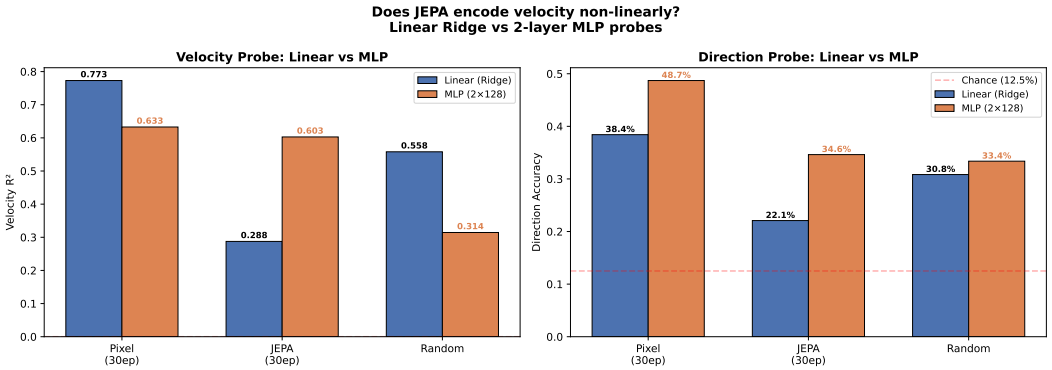

Figure 16: **Non-linear velocity encoding in JEPA-style representations.** Linear probes (left bars) versus MLP probes (right bars) on frozen BeliefField features. The JEPA-style model is the *only* one whose velocity R^2 *increases* with a non-linear probe (from 0.29 to 0.60, $\Delta = +0.31$), while Pixel and Random both *decrease*. This confirms that the PDE with a JEPA-style objective encodes dynamics in a distributed, non-linearly accessible form, consistent with abstract representations. Linear probes systematically underestimate the JEPA-style model's encoding of physical dynamics.

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A Hyperparameter Details

Table 7: **Full hyperparameter specification.**

Hyperparameter	FLUIDWORLD	Transformer	ConvLSTM
Architecture			
Latent dimension d	128	128	128
Patch size p	4	4	4
Spatial resolution $H_f \times W_f$	16×16	16×16	16×16
Encoder layers	3 (FluidLayer2D)	2 (TransformerBlock)	Conv bottleneck
Encoder mid channels	—	—	88
Temporal layers	3 (BeliefField evolve)	1 (TransformerBlock)	1 (ConvLSTMCell)
ConvLSTM hidden channels	—	—	64
ConvLSTM kernel size	—	—	3
Attention heads	—	8	—
FFN dimension	$2d = 256$ (reaction)	384	—
Dilations	$\{1, 4, 16\}$	—	—
Max PDE steps (encoder)	6	—	—
Max PDE steps (BeliefField)	3	—	—
PDE Δt (initial)	0.1 (learned)	—	—
PDE ϵ (stopping)	0.08	—	—
Positional embeddings	—	Learned (256×128)	—
Decoder mid channels	64	64	64
Bio mechanisms (FluidWorld only)			
Lateral inhibition β	0.3	—	—
Fatigue cost κ	0.1	—	—
Fatigue recovery ρ	0.02	—	—
Hebbian decay λ	0.99	—	—
Hebbian LR η	0.01	—	—
Hebbian gain α_H	0.5	—	—
BeliefField decay γ	0.95 (learned)	—	—
Training			
Optimizer	AdamW	AdamW	AdamW
Learning rate	3×10^{-4}	3×10^{-4}	3×10^{-4}
Weight decay	0.04	0.04	0.04
Batch size	16	16	16
BPTT window T	4	4	4
LR warmup steps	500	500	500
LR schedule	Cosine annealing	Cosine annealing	Cosine annealing
Grad clip norm	1.0	1.0	1.0
Precision	FP16 (AMP)	FP16 (AMP)	FP16 (AMP)
Loss weights			
w_r (reconstruction)	1.0	1.0	1.0
w_p (prediction)	1.0	1.0	1.0
w_v (variance)	0.5	0.5	0.5
w_g (gradient)	1.0	1.0	1.0
σ_{target} (var target)	1.0	1.0	1.0

B Architectural Details

B.1 FluidLayer2D (PDE Integration Step)

Algorithm 1 details the full integration loop of one FluidLayer2D module.

Algorithm 1 FluidLayer2D Forward Pass

Require: Input feature map $u \in \mathbb{R}^{B\times d\times H\times W}$
Ensure: Output feature map $u' \in \mathbb{R}^{B\times d\times H\times W}$, diagnostics
Initialize $h_g \leftarrow 0$, $h_l \leftarrow 0$, $\tau_{\text{stopped}} \leftarrow$ `max_steps`
for $\tau = 1$ to `max_steps` **do**
  diff $\leftarrow \sum_k \text{softplus}(\hat{D}_k) \cdot \text{Conv2D}(u, K_{\text{lap}}, d_k)$ {Multi-scale Laplacian}
  react $\leftarrow \text{MLP}(u)$ {Position-wise reaction}
  $\bar{u} \leftarrow$ spatial_mean(u)
  $h_g \leftarrow h_g + \text{sigmoid}(W_g\bar{u}) \cdot \tanh(W_v\bar{u})$ {Global memory}
  $h_l \leftarrow$ update_local(u) {Local 4×4 memory}
  $\Delta t \leftarrow \exp(\widehat{\Delta t})$ clamped to $[0.005, 0.35]$
  $u \leftarrow u + \Delta t \cdot (\text{diff} + \text{react} + \alpha_g h_g + \alpha_l h_l)$
  if $\tau \bmod 2 = 0$ **then**
    $u \leftarrow \text{RMSNorm}(u)$
  end if
  if eval mode **and** adaptive stopping criterion (Eq. 5) met **then**
    $\tau_{\text{stopped}} \leftarrow \tau$; **break**
  end if
end for
return u, $\{\text{steps_used} : \tau_{\text{stopped}}, \text{turbulence}, \text{energy}\}$

B.2 Decoder Architecture

The PixelDecoder follows a symmetric upsampling path:

Layer	Output Shape	Details
Input	$(B, 128, 16, 16)$	Latent features
Conv 1×1	$(B, 64, 16, 16)$	Channel projection
ResBlock	$(B, 64, 16, 16)$	3×3 Conv $\times2$ + skip
Upsample $\times2$ + Conv 3×3	$(B, 64, 32, 32)$	Bilinear + Conv
GroupNorm + ResBlock	$(B, 64, 32, 32)$	Mid-resolution detail
Upsample $\times2$ + Conv 3×3	$(B, 32, 64, 64)$	Bilinear + Conv
GroupNorm + ResBlock	$(B, 32, 64, 64)$	Fine detail
Conv 3×3	$(B, C, 64, 64)$	Output logits

Total decoder parameters: $\sim$231K. Bilinear upsampling avoids checkerboard artifacts [26].

C Additional Experimental Details

Dataset. UCF-101 [32] contains 13,320 videos across 101 human action categories. I resize all frames to 64×64 and extract temporal windows of $T + 1 = 5$ consecutive frames. No data augmentation is applied to ensure fair comparison.

Convergence dynamics. Figure 17 shows how the representation quality gap evolves during training.

Autoregressive rollouts. All three models produce recognizable content at $h = 1$ (one-step prediction). Beyond $h = 1$, all degrade, but with distinct failure patterns and at different rates:

- **PDE:** Slowest degradation. Gradual contrast reduction, color shifts toward dark/green tones. Spatial structure (edges, object boundaries) is preserved up to $h=3$ due to the Laplacian's inherent spatial smoothing.
- **Transformer:** Convergence toward spatial mean color (brown/orange uniform patches). Global attention averages away spatial detail.

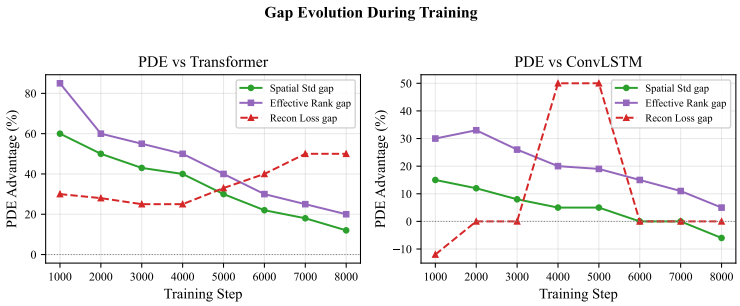

Figure 17: **Gap evolution during training.** Spatial Std and Effective Rank advantages narrow as the Transformer learns spatial patterns from data, but reconstruction fidelity gap *widens*. The PDE advantage is structural, not merely a convergence artifact.

- **ConvLSTM:** Dissolution into repetitive texture artifacts (green/brown noise patterns). The fixed 3×3 receptive field cannot propagate global context, and the LSTM state accumulates spatially incoherent errors. Despite having spatial inductive bias (convolutions), the ConvLSTM degrades as fast as the Transformer, demonstrating that spatial bias alone is insufficient without the PDE's continuous diffusion mechanism.

These distinct failure modes reflect the different inductive biases: the PDE preserves spatial relationships longest due to the multi-scale Laplacian acting as an implicit spatial regularizer at each autoregressive step.

D Supplementary Figures from Experimental Notebooks

The following figures are produced by the experimental notebooks and provide visual evidence for claims made in the main text.

D.1 Symmetry Breaking (cf. §7)

D.2 Phase Diagram and Energy Landscape (cf. §7)

D.3 Energy Conservation and Numerical Stability (cf. §7)

D.4 Resilience and Self-Repair (cf. §7)

D.5 Persistent Memory via Titans (cf. §7)

D.6 BeliefField Evolution (cf. §4.4)

D.7 Feature Analysis (cf. §5)

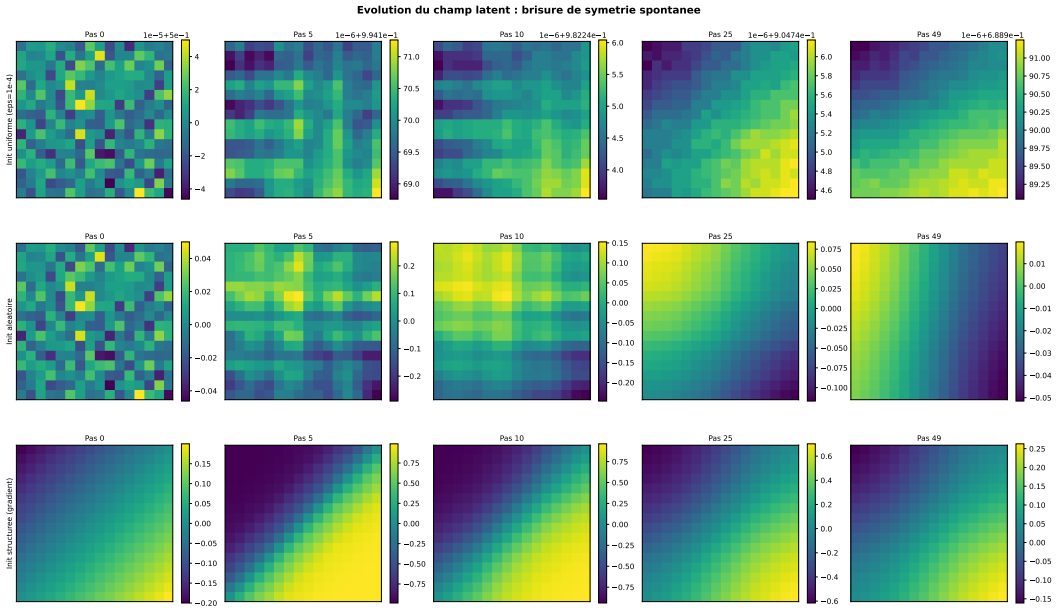

Figure 18: Spatial heatmaps of the BeliefField at successive integration steps, starting from a near-uniform initialization ($\epsilon = 10^{-4}$). Within 10 steps, distinct spatial clusters emerge deterministically. The process is sensitive to initial conditions (different seeds produce different patterns) but fully reproducible given identical seeds.

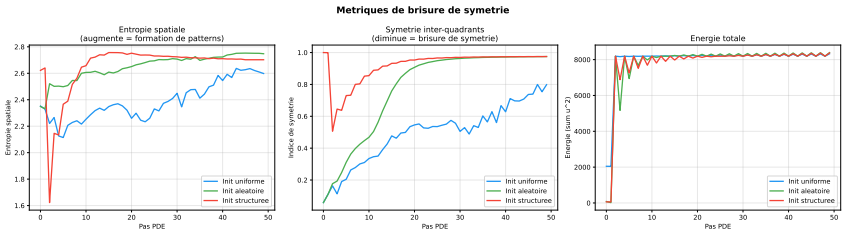

Figure 19: Quantitative symmetry-breaking metrics. Symmetry index drops from 1.0 to 0.2, spatial entropy rises from 1.7 to 2.7, and KMeans detects 3 emerging clusters. All transitions occur within the first 10 integration steps.

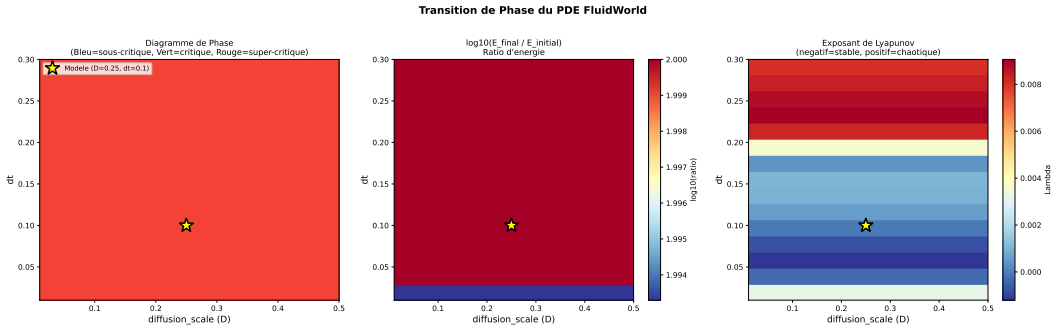

Figure 20: Phase diagram across 225 $(D, \Delta t)$ configurations. The trained operating point ($D \approx 0.25$, $\Delta t = 0.1$) is firmly in the super-critical regime. The only sub-critical region occurs at $\Delta t \approx 0.02$, where dynamics are effectively frozen.

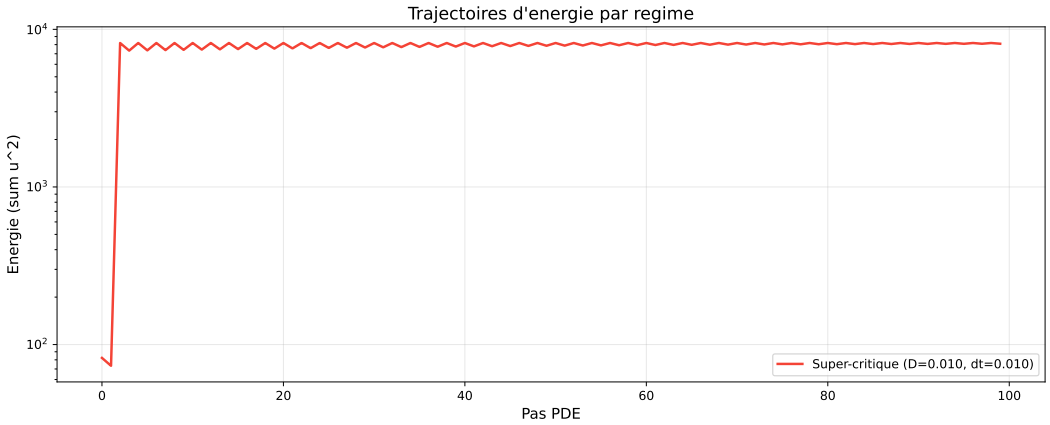

Figure 21: Energy trajectories from multiple initial conditions. Regardless of initialization (uniform, random, structured gradient), energy converges to the same attractor $E^* \approx 8{,}640$ within $\sim$50 steps, confirming bounded attractor dynamics.

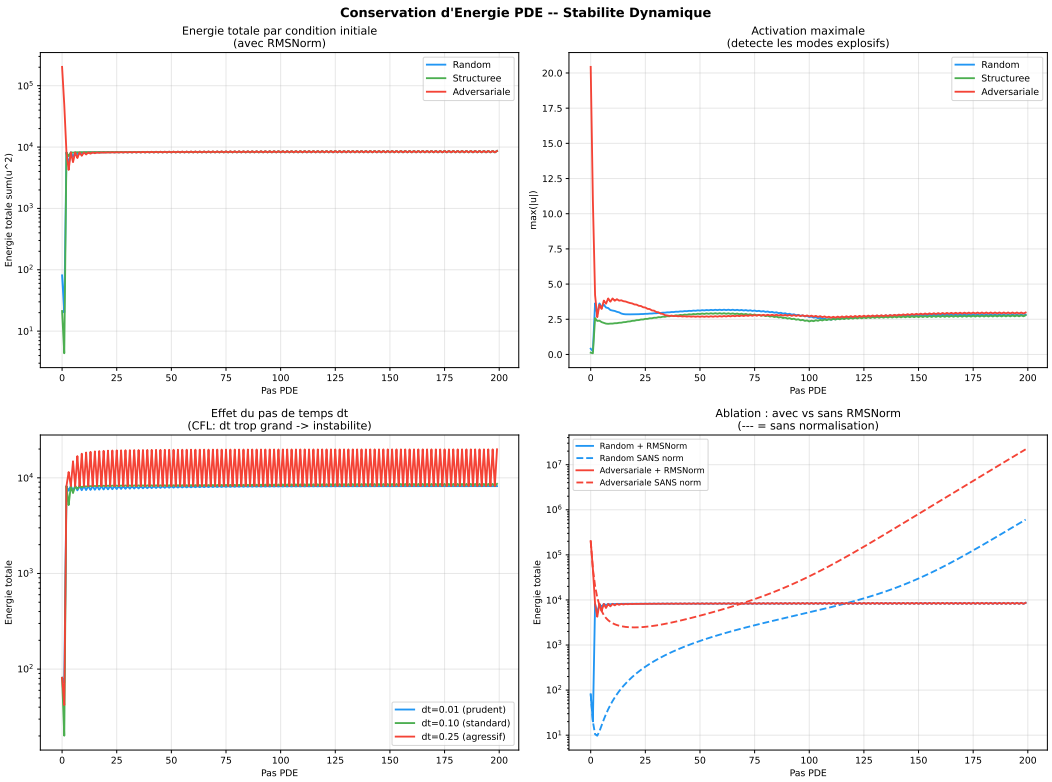

Figure 22: Four-panel energy conservation analysis. The PDE substrate with RMSNorm maintains bounded energy; without RMSNorm, energy diverges exponentially ($\times 7{,}467$ over 200 steps). High-frequency components decay to 10^{-8} by step 200.

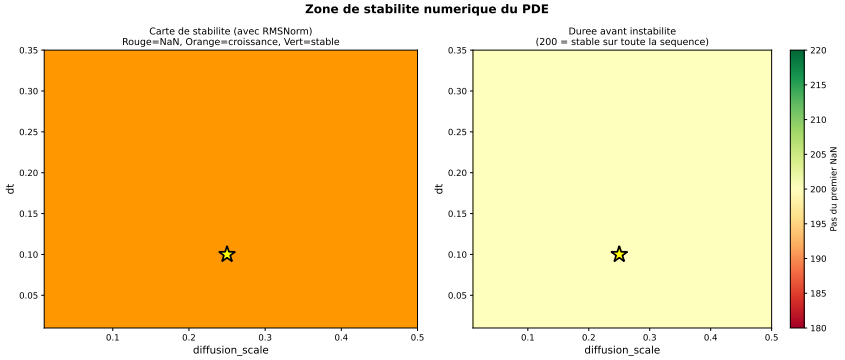

Figure 23: Numerical stability across integration timesteps. Oscillatory instabilities appear when $\Delta t > 0.10$, confirming the CFL-like stability bound for the discrete Laplacian operator.

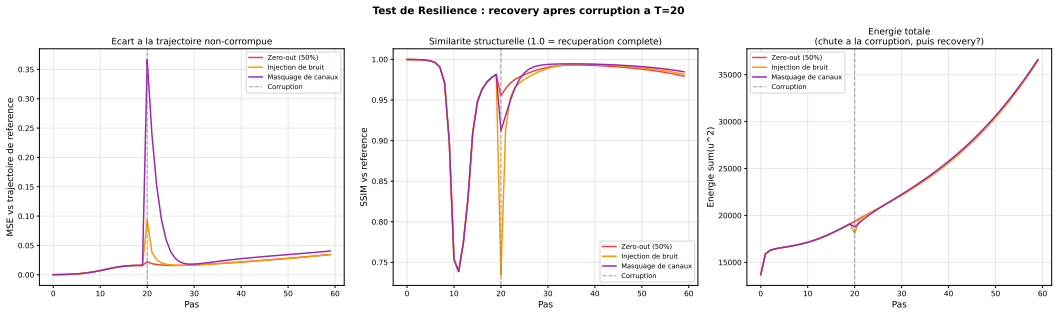

Figure 24: Recovery dynamics after 50% corruption of the BeliefField. Three modalities are shown: channel zeroing (instant recovery), Gaussian noise (3-step recovery), and spatial masking (7-step recovery). In all cases the system returns to its pre-corruption trajectory.

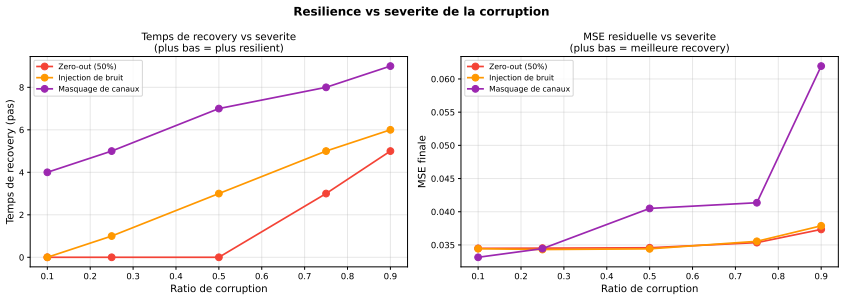

Figure 25: Resilience across corruption intensities (10%–90%). Residual MSE remains flat (≈ 0.034) up to 50% corruption, then increases monotonically but without discontinuity. No "cliff effect" at any corruption level.

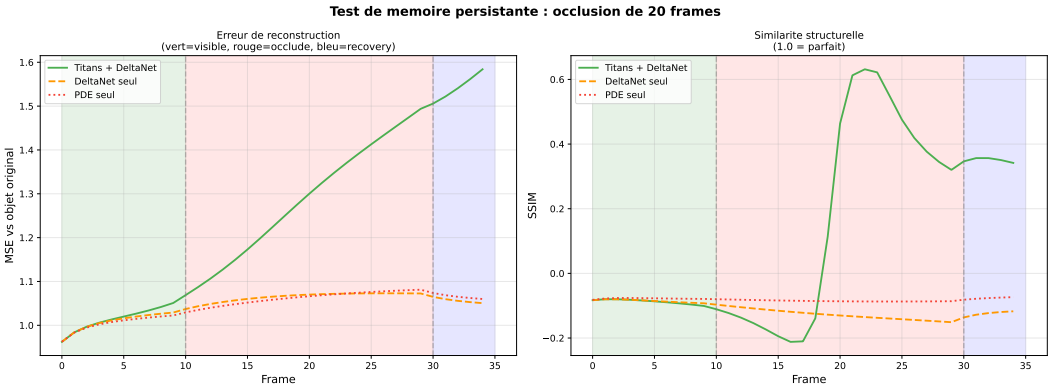

Figure 26: Memory persistence test. An object is shown for 10 frames, occluded for 20 frames, then revealed. Titans-augmented model achieves SSIM recovery of 0.6 (vs 0.05 for PDE-only). Memory activity $|M|$ increases during occlusion (0.018 to 0.035), indicating active information retention despite absent visual input.

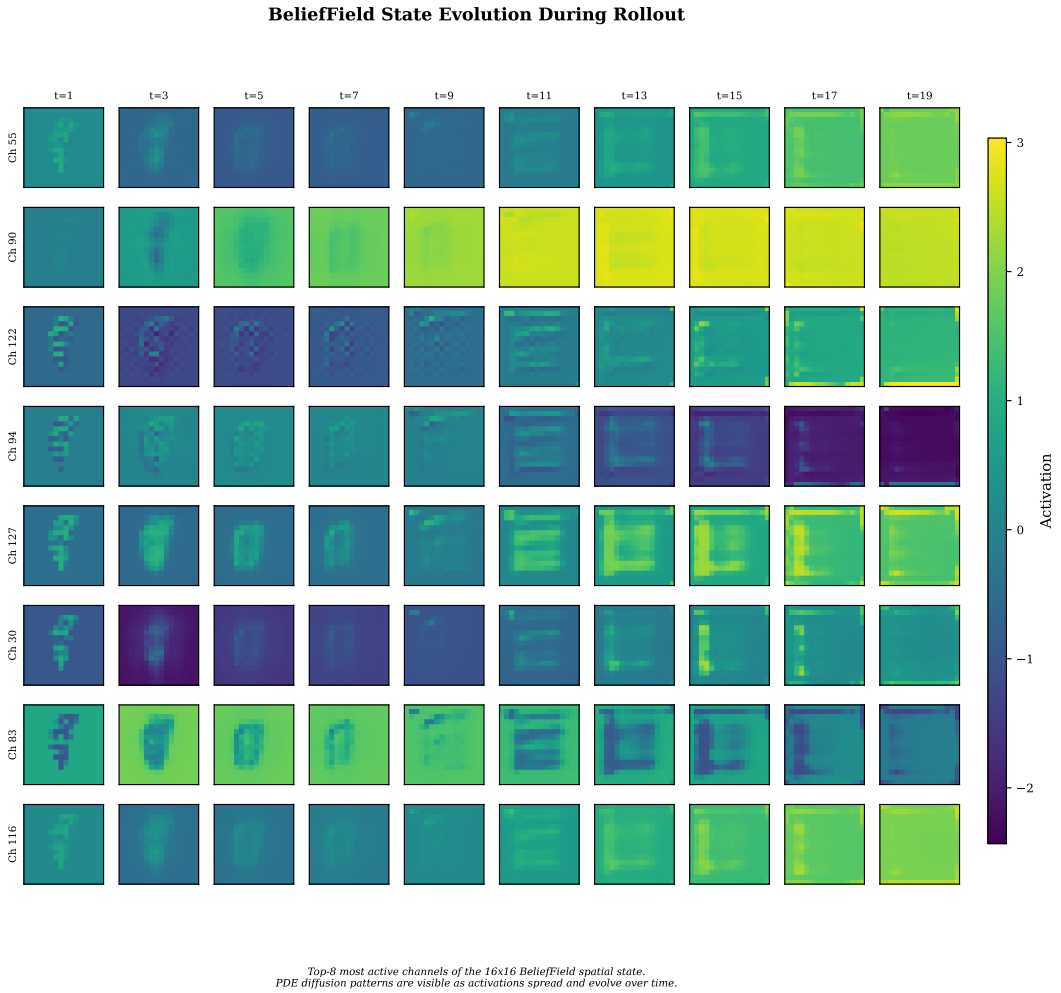

Figure 27: Visualization of the BeliefField's internal state during temporal evolution. The spatial field accumulates structure from successive frames, forming a persistent predictive representation that carries forward through PDE integration.

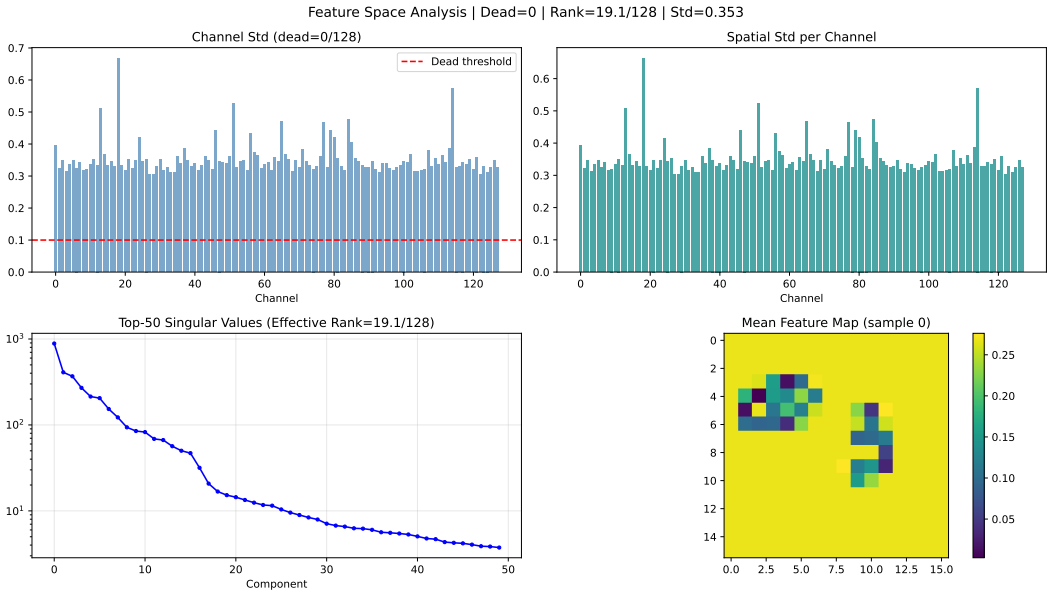

Figure 28: Feature-space analysis of the pixel-trained model. Spatial feature maps show clear semantic encoding of object positions and boundaries. 0 of 128 channels are dead, effective rank is 19.5, confirming healthy representational diversity.